

Plants aren't dumb

Using optimality theory to understand big questions in plant ecophysiology

Nick Smith

Department of Biological Sciences

Texas Tech University

nick.smith@ttu.edu

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Using optimality theory to understand **a big question** in plant ecophysiology

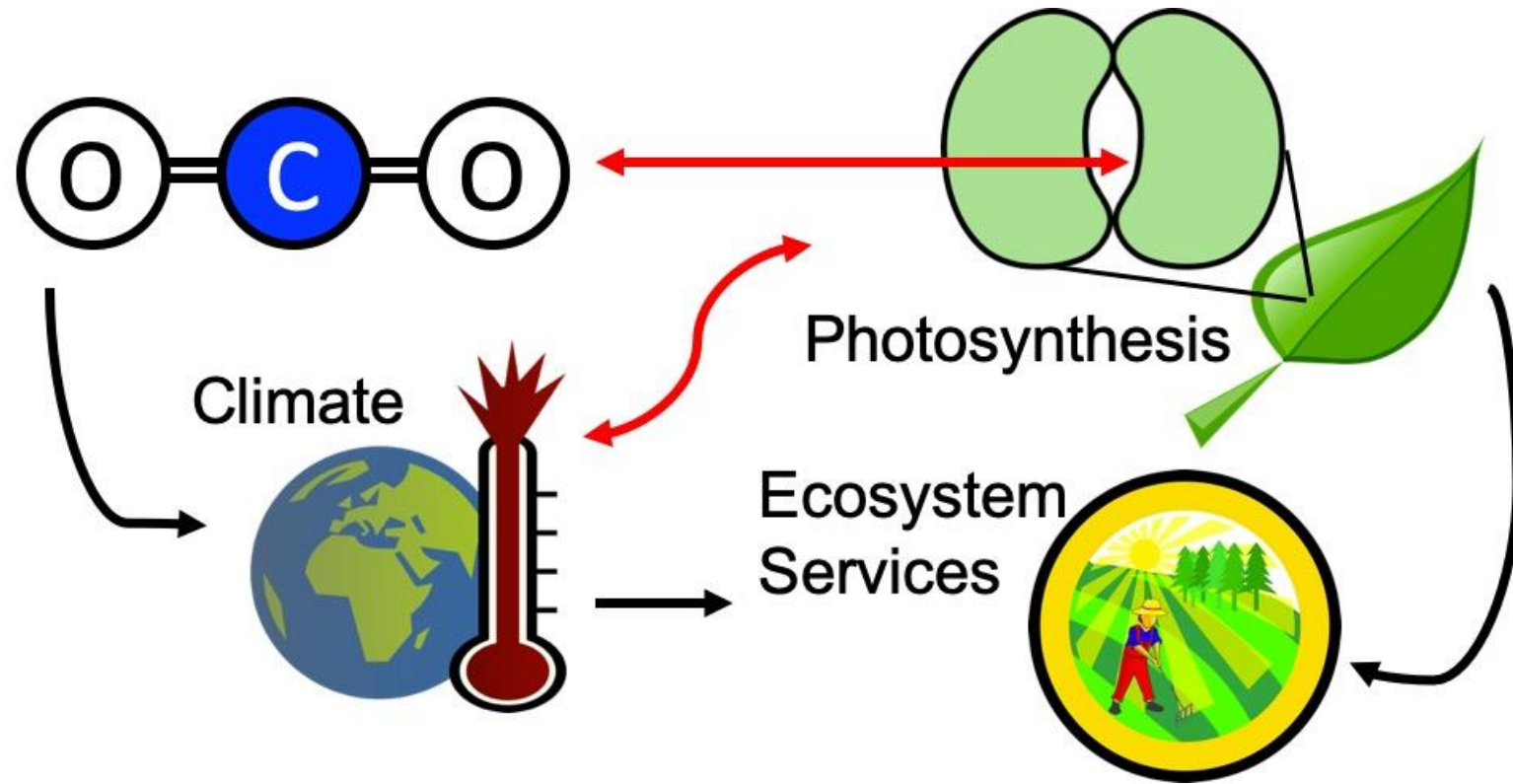
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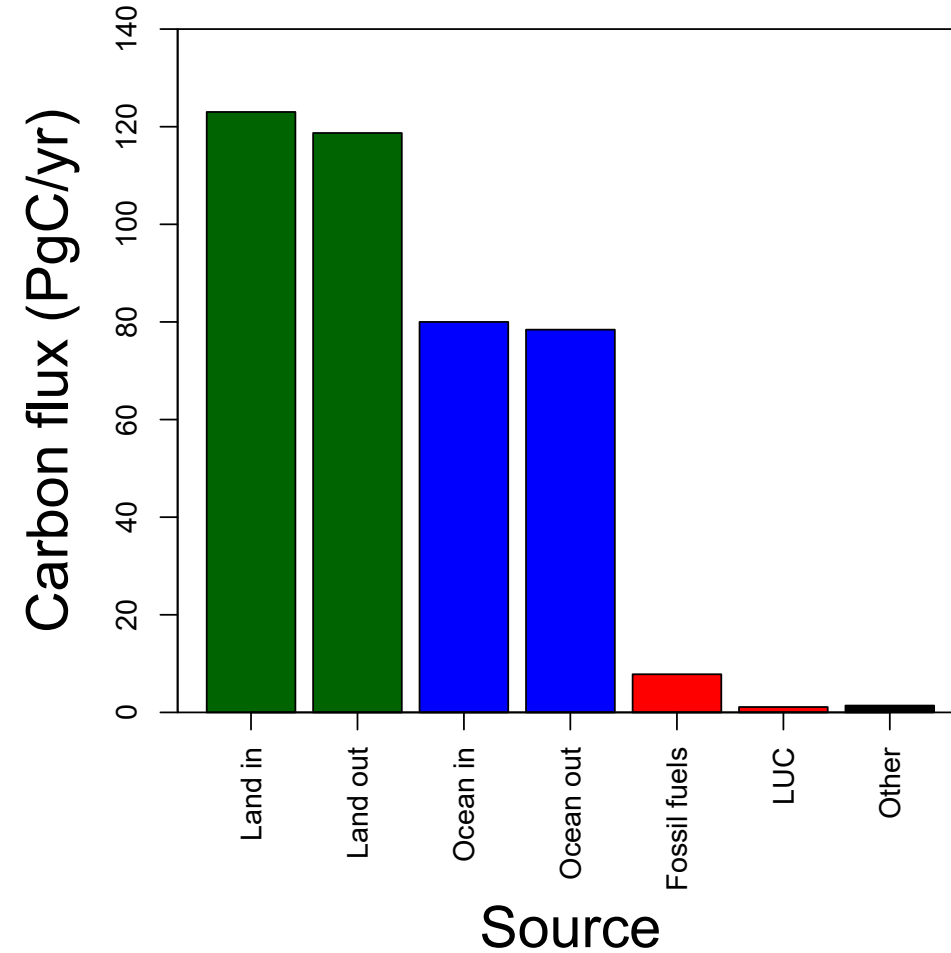
nick.smith@ttu.edu

Our lab examines **photosynthesis** as a regulator of global change impacts on ecosystem services

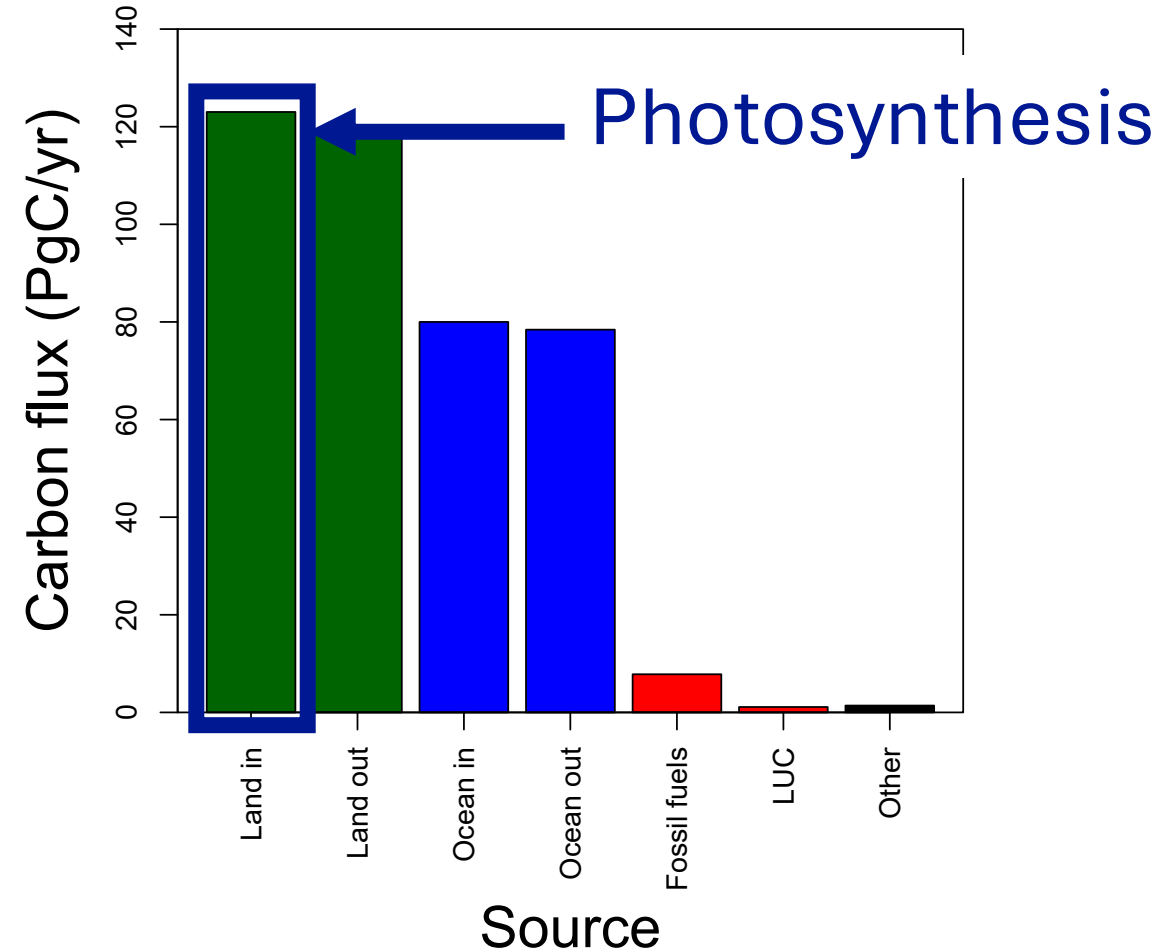


Photosynthesis is important! An example...

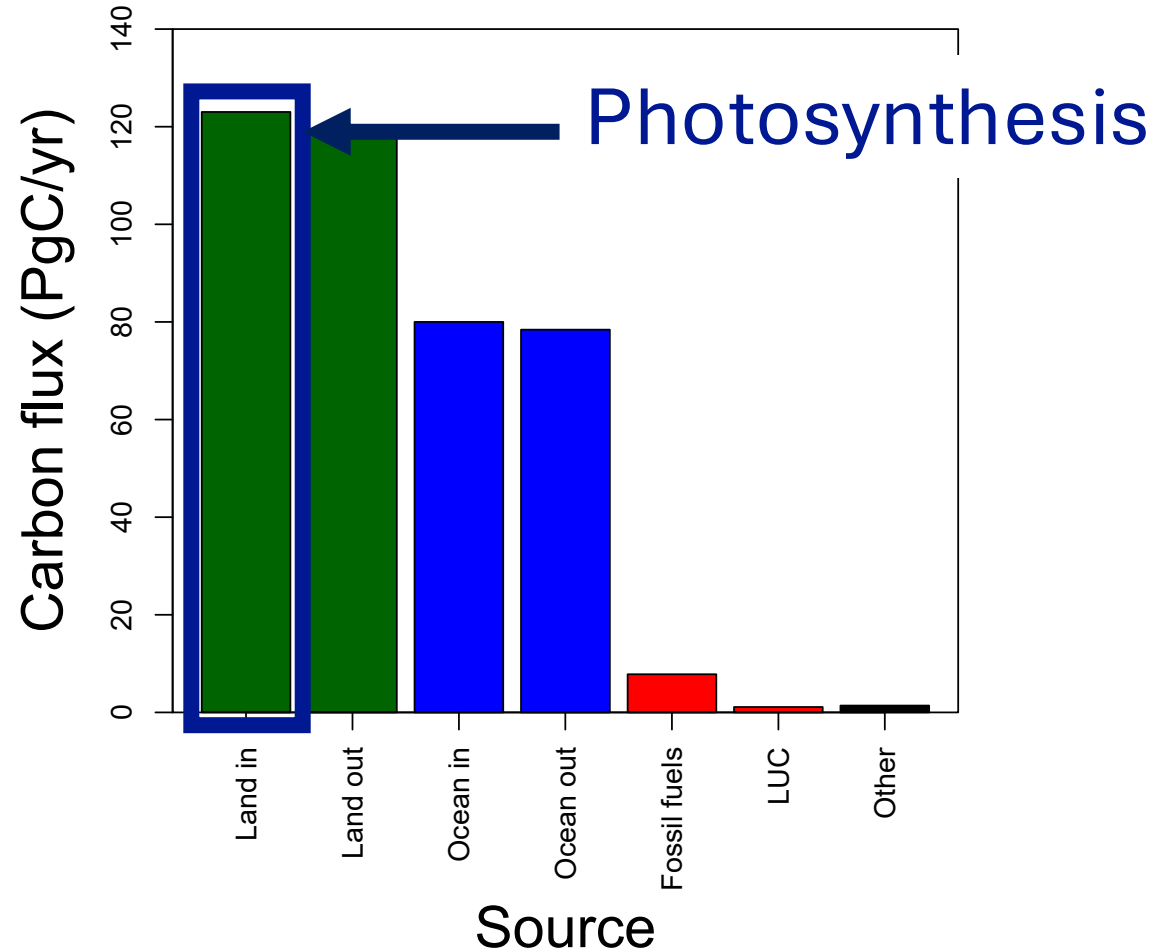
Photosynthesis is important! An example...



Photosynthesis is important! An example...

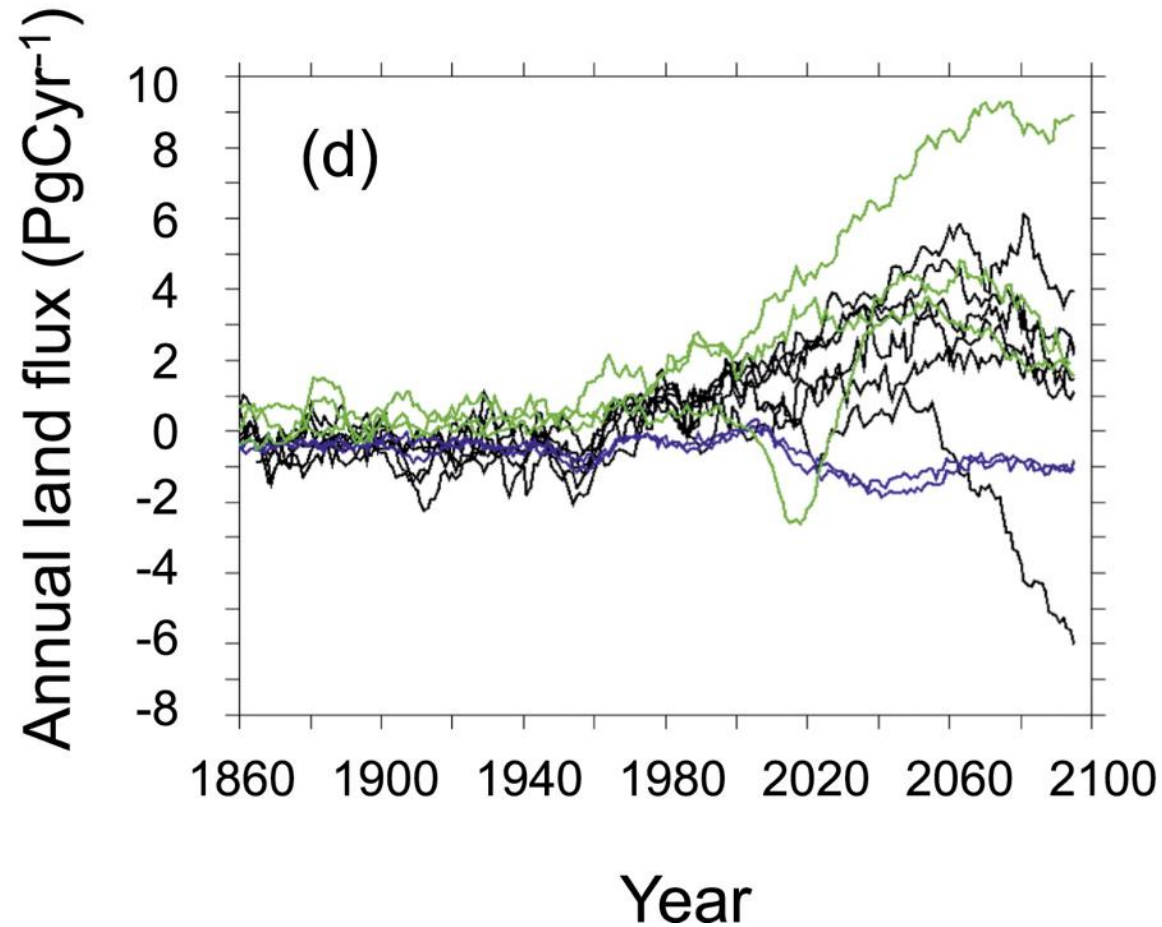


Photosynthesis is important! An example...

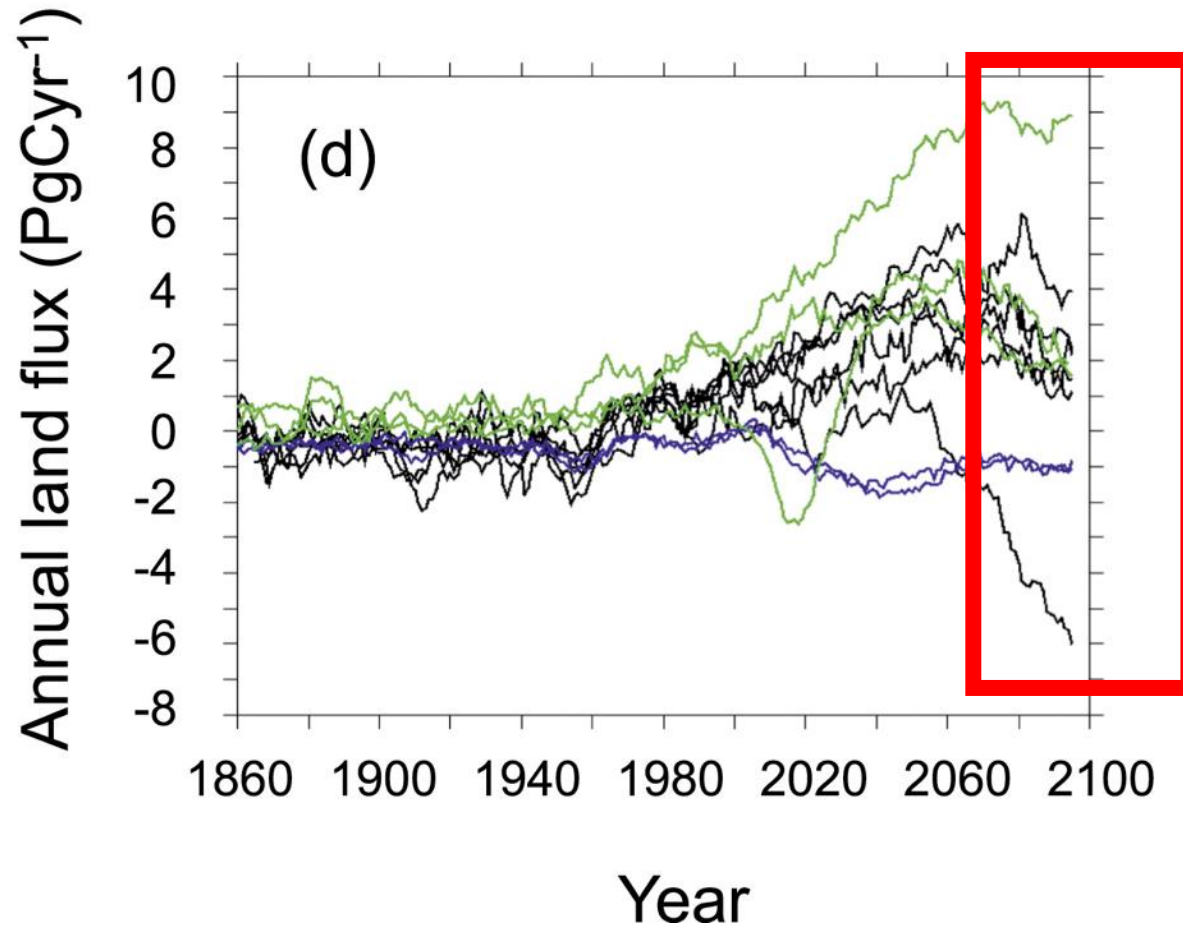


A small percentage change in photosynthesis can have large consequences for climate

But predictions are uncertain!

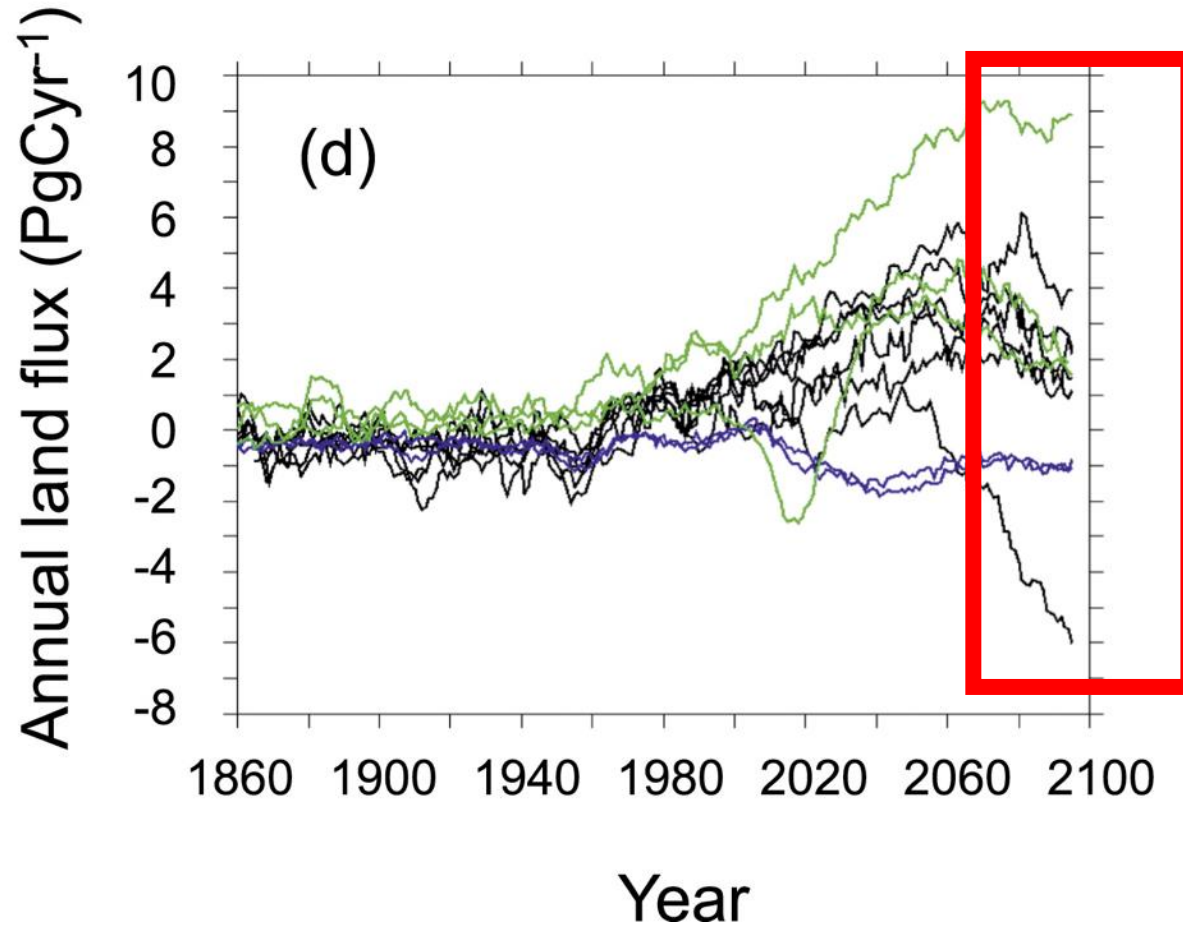


But predictions are uncertain!

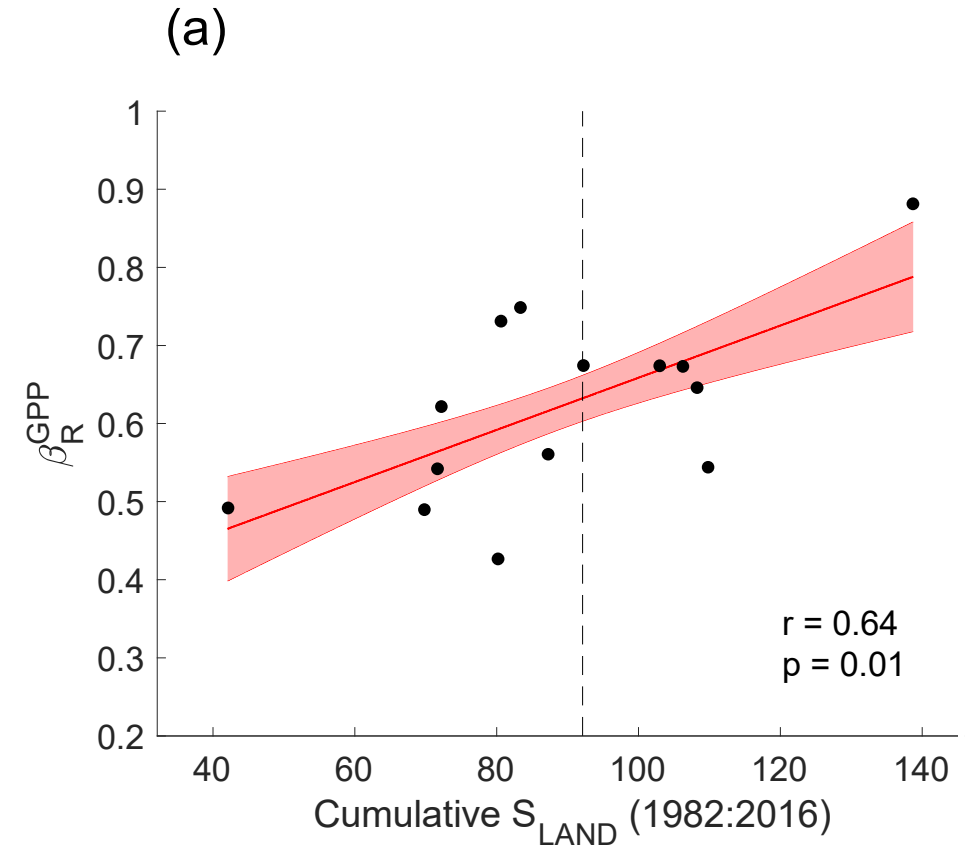


Future model uncertainty (14 Pg) > current fossil fuel emissions (9.5 Pg)

This uncertainty is driven by uncertainty in photosynthesis



Friedlingstein et al. (2014) *J Climate*



Keenan et al. (2023) *Nature Climate Change*

Why the uncertainty? Theoretical models for photosynthesis exist

Planta 149, 78–90 (1980)

Planta
© by Springer-Verlag 1980

A Biochemical Model of Photosynthetic CO₂ Assimilation in Leaves of C₃ Species

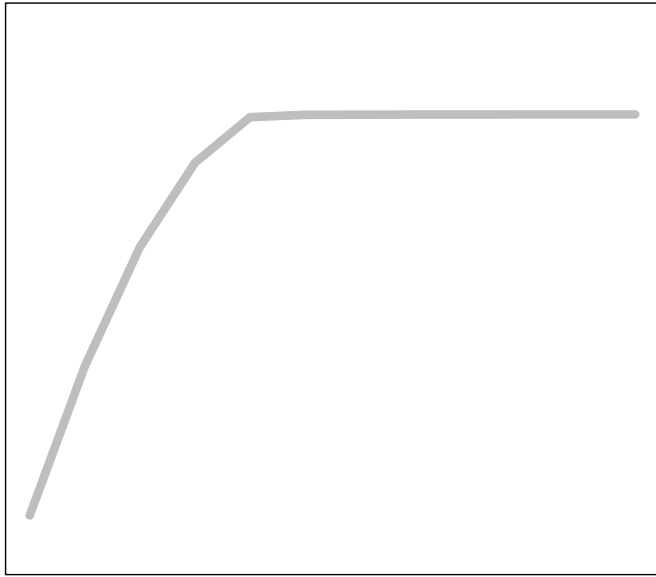
G.D. Farquhar¹, S. von Caemmerer¹, and J.A. Berry²

¹ Department of Environmental Biology, Research School of Biological Sciences, Australian National University, P.O. Box 475, Canberra City ACT 2601, Australia and

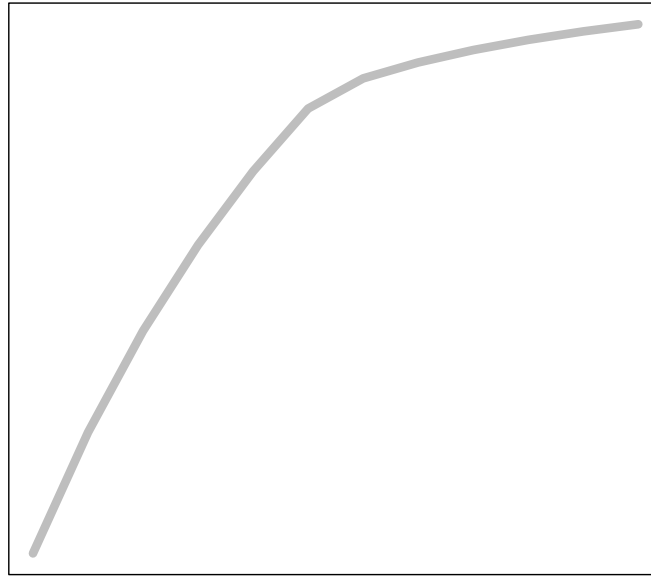
² Carnegie Institution of Washington, Department of Plant Biology, Stanford, Cal. 94305, USA

These produce short term responses that match data

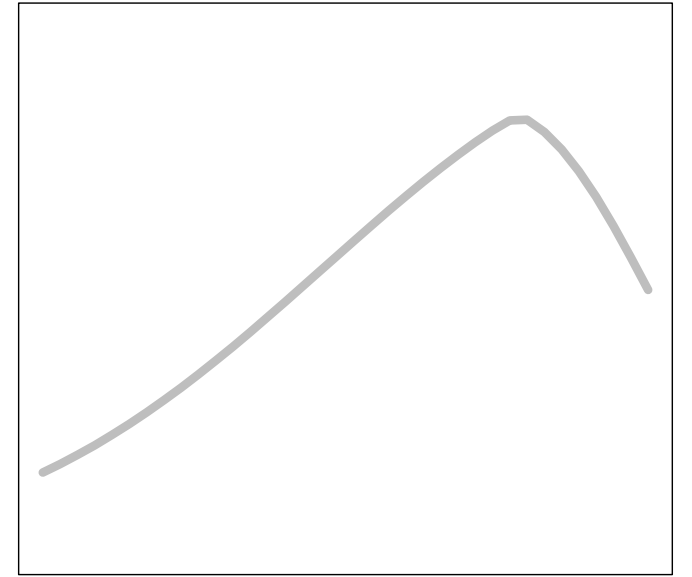
Photosynthetic rate



Light

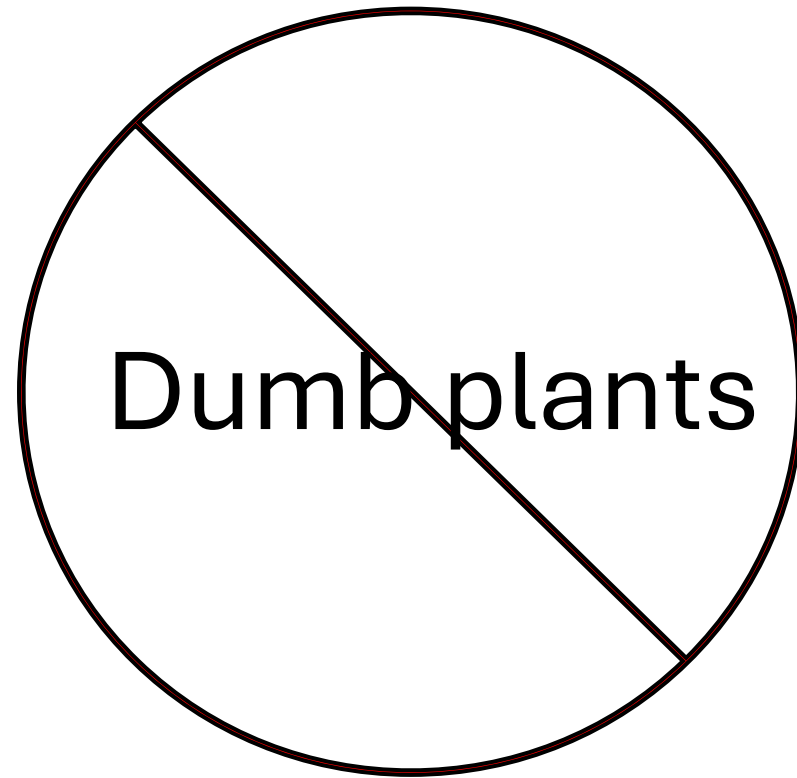


CO₂



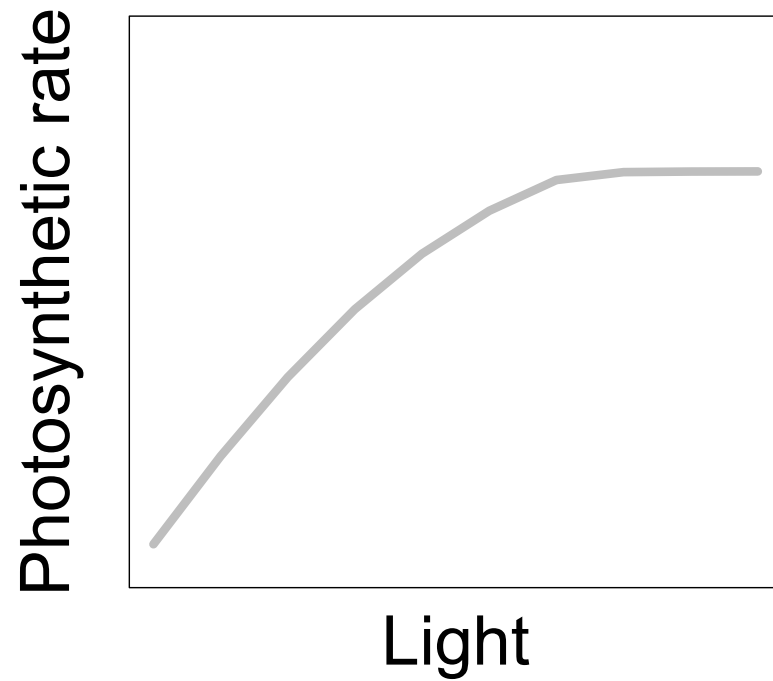
Temperature

Why the uncertainty? Long-term responses differ from short-term responses due to acclimation

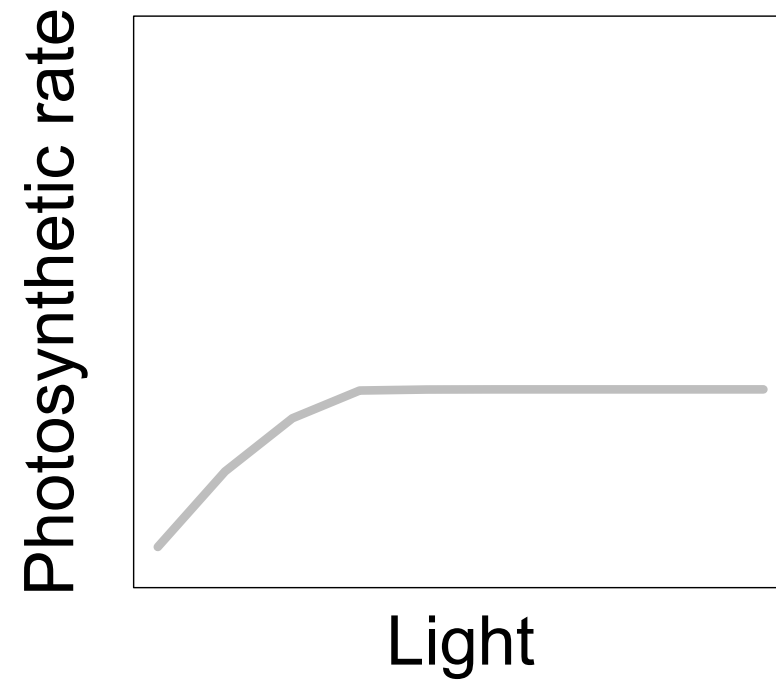


Acclimation: an example

Acclimated to high light

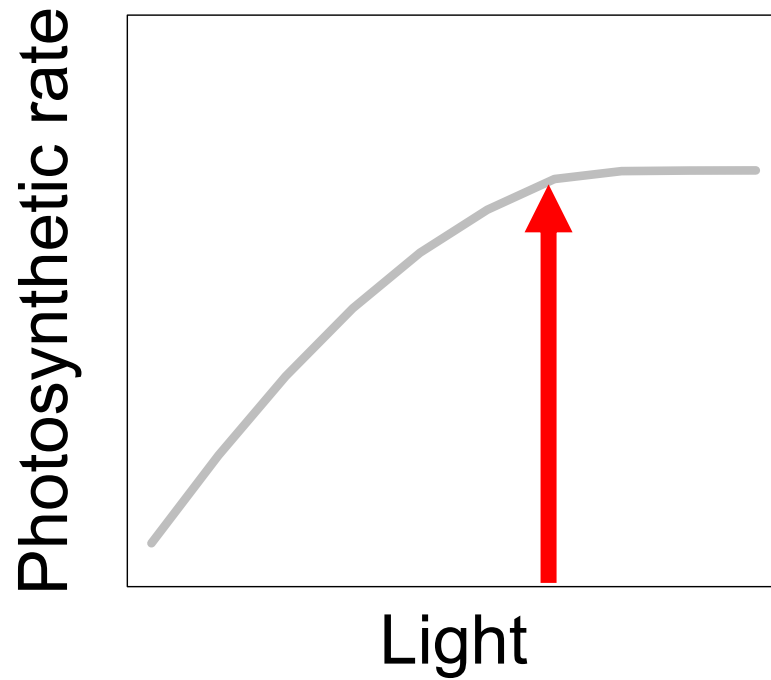


Acclimated to low light

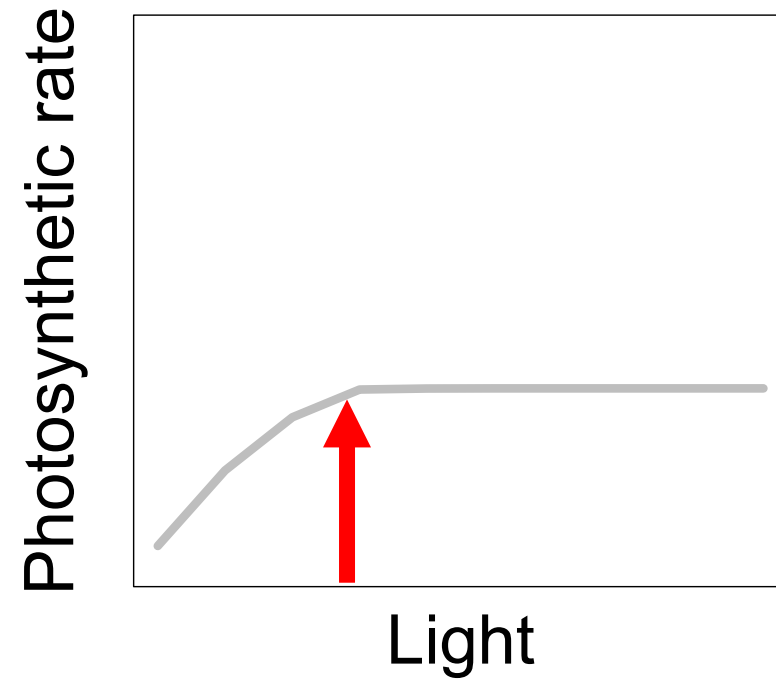


Acclimation: an example

Acclimated to high light



Acclimated to low light



Photosynthesis peaks a light level close to the level the leaf is acclimated to

Acclimation is ubiquitous and well known...

CO₂: Bazzaz (1990)

Annu. Rev. Ecol. Syst. 1990. 21:167–96
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THE RESPONSE OF NATURAL
ECOSYSTEMS TO THE RISING
GLOBAL CO₂ LEVELS

F. A. Bazzaz

Light: Boardman (1977)

Ann. Rev. Plant Physiol. 1977. 28:355–77
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COMPARATIVE
PHOTOSYNTHESIS OF SUN
AND SHADE PLANTS

N. K. Boardman

Division of Plant Industry, CSIRO, Canberra City, A.C.T. 2601, Australia

Temperature: Berry &
Björkman (1980)

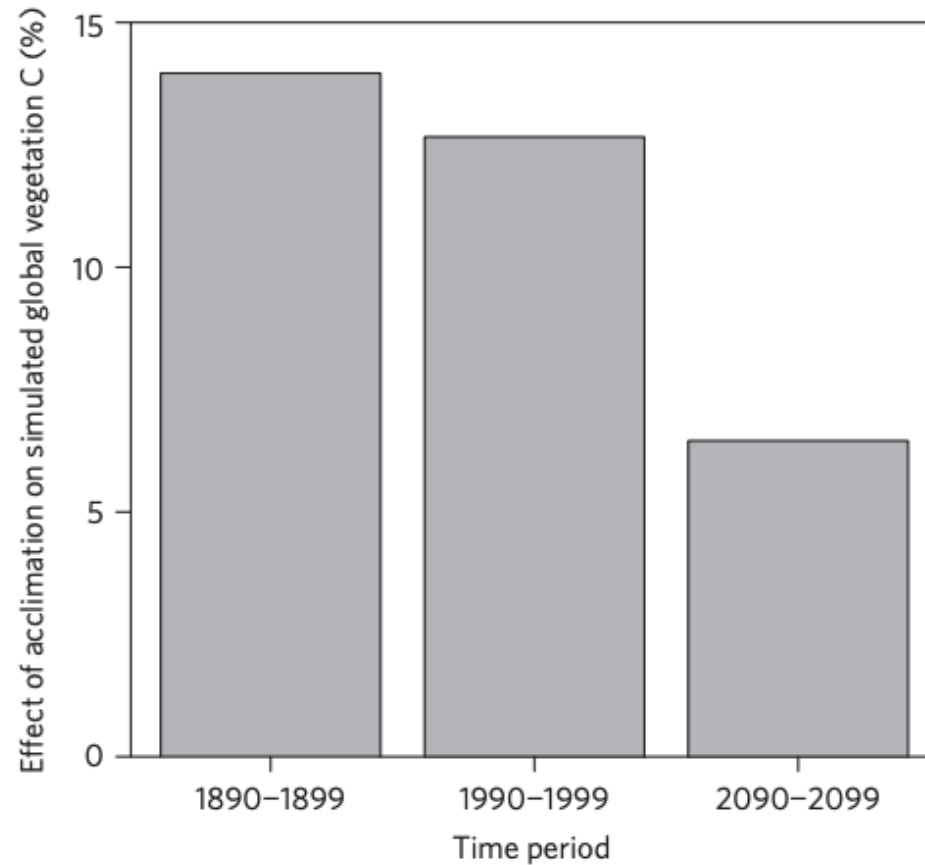
Ann. Rev. Plant Physiol. 1980. 31:491–543
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PHOTOSYNTHETIC RESPONSE
AND ADAPTATION
TO TEMPERATURE
IN HIGHER PLANTS

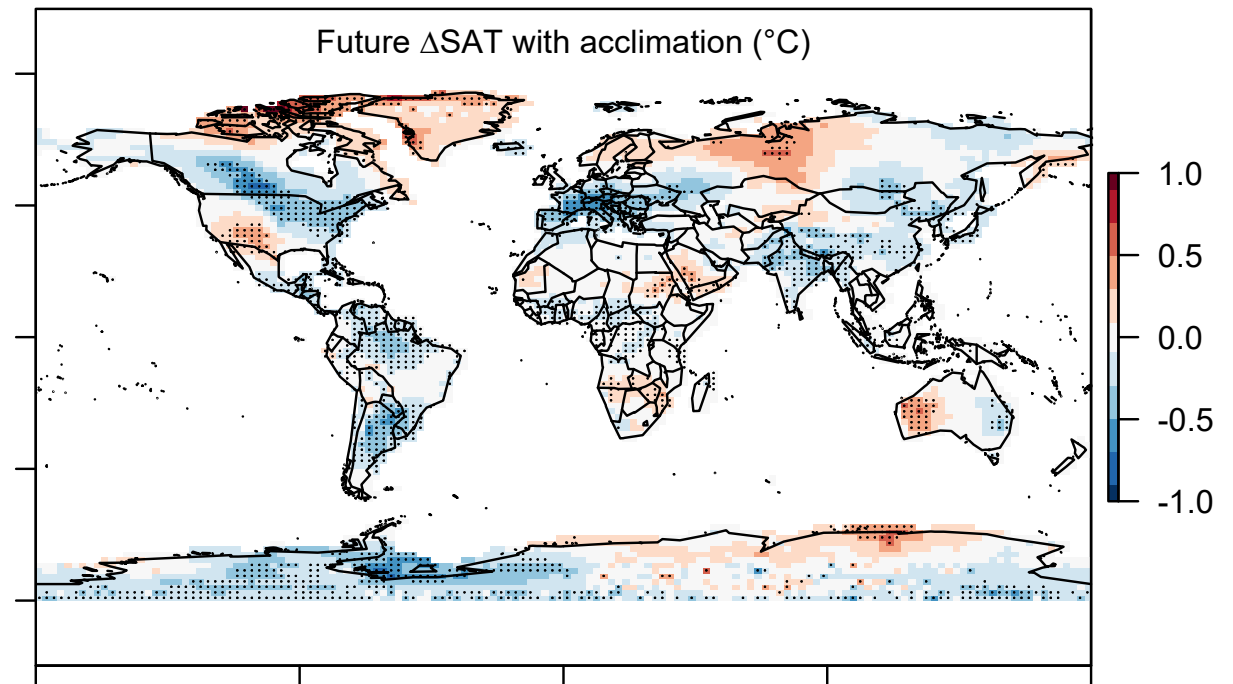
- *Joseph Berry and Olle Björkman*¹

...and can impact carbon cycling and climate

Acclimation increase future C storage by ~6%



Acclimation alters future temperature by >1°C

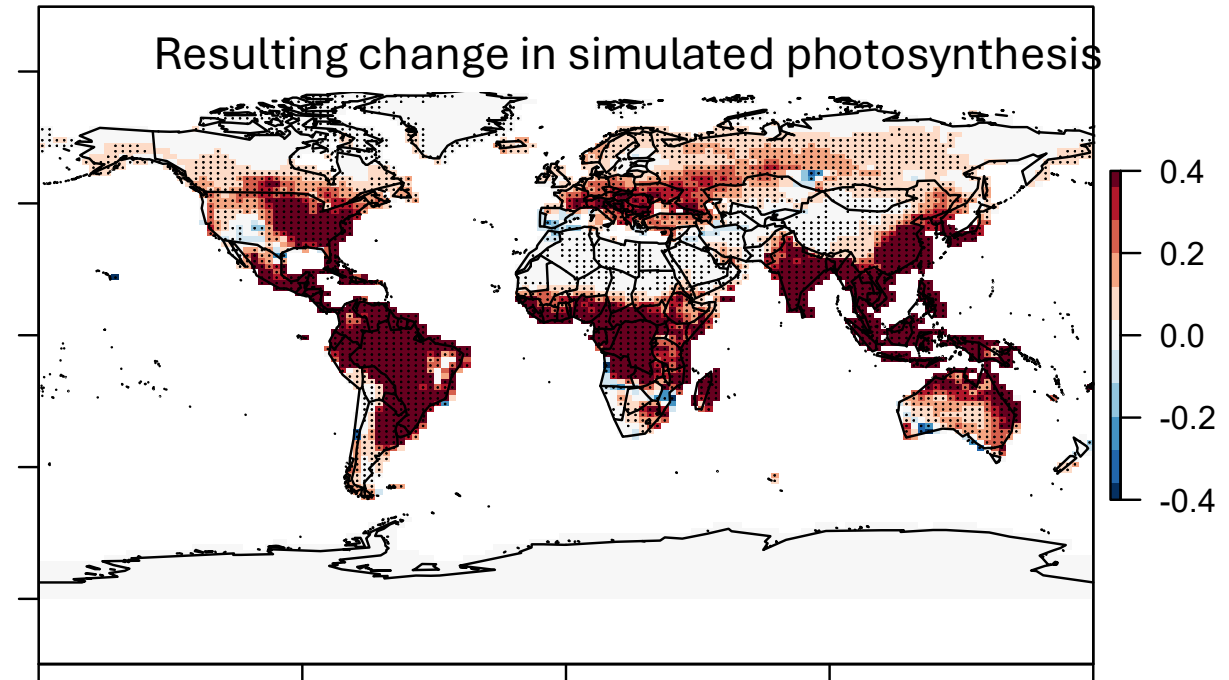
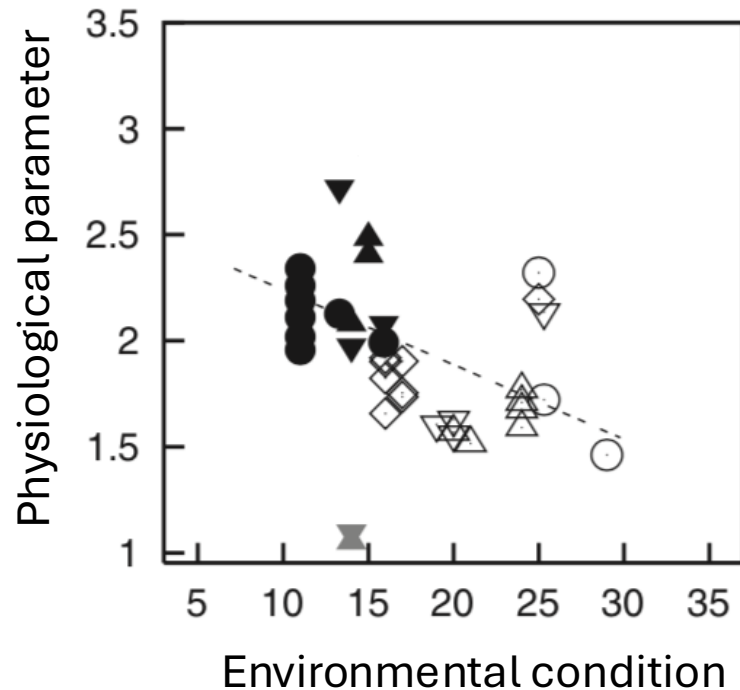


Unfortunately,
no theoretical model for
photosynthetic acclimation exists

Lack of theory results in...

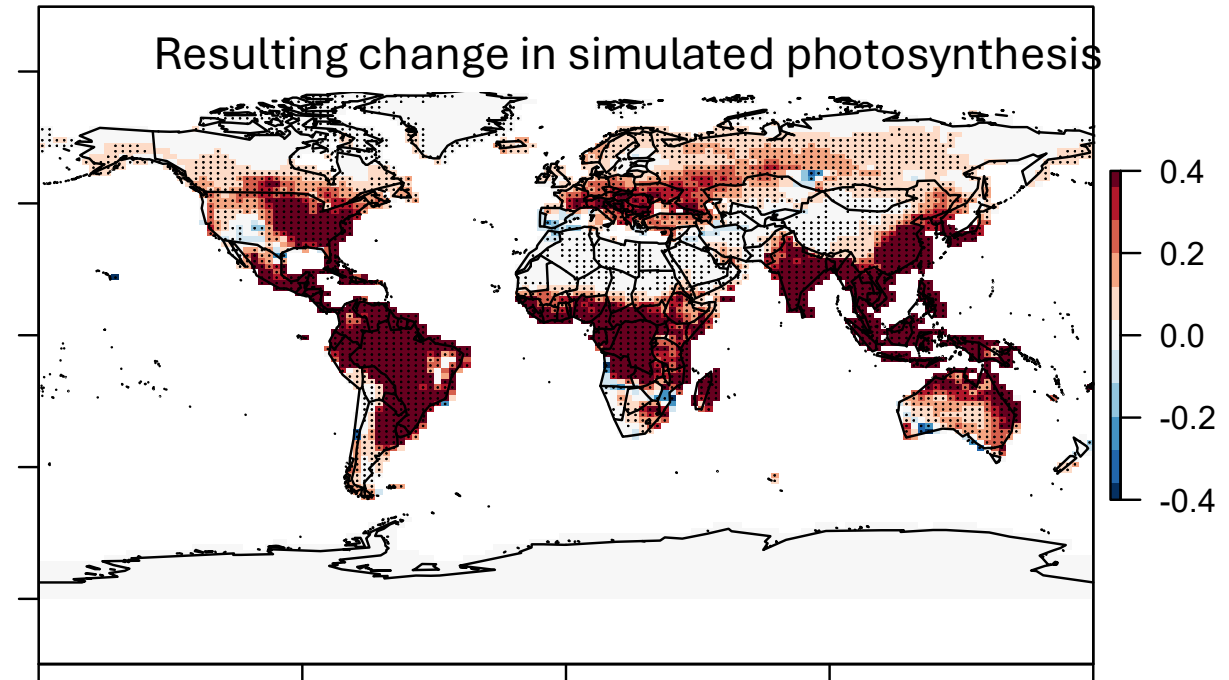
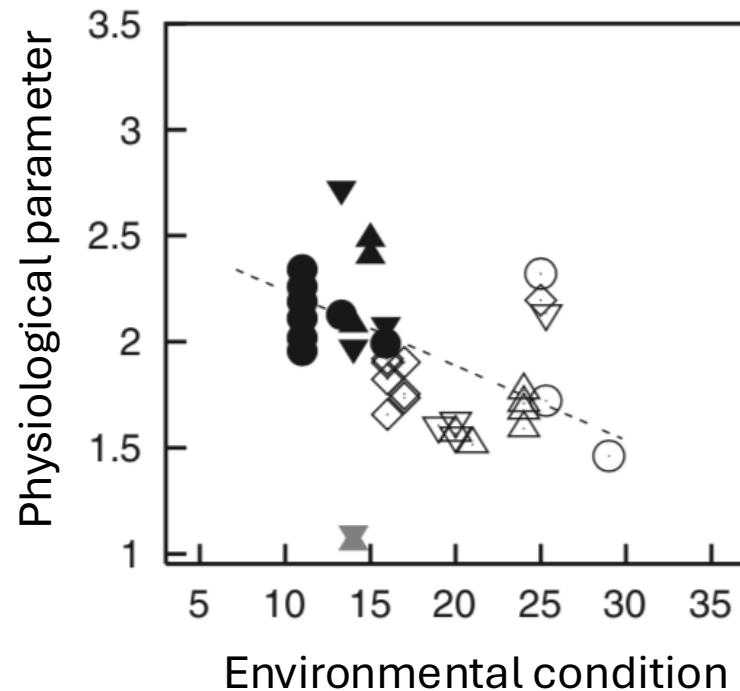
Lack of theory results in...

- Unreliable future predictions (overparameterization, tuning)
 - Reliance on statistical models



Lack of theory results in...

- Inability to test mechanisms



La

• In



Global Change Biology

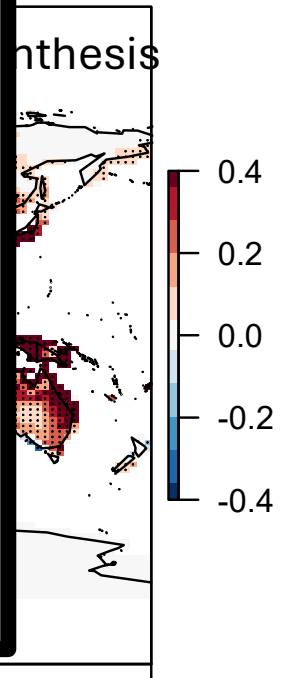
REPORT | Open Access |

Increasing the spatial and temporal impact of ecological research: A roadmap for integrating a novel terrestrial process into an Earth system model

Emily Kyker-Snowman , Danica L. Lombardozzi, Gordon B. Bonan, Susan J. Cheng, Jeffrey S. Dukes, Serita D. Frey, Elin M. Jacobs, Risa McNellis, Joshua M. Rady, Nicholas G. Smith ... [See all authors](#) ▾

First published: 20 September 2021 | <https://doi.org/10.1111/gcb.15894> | Citations: 27

See also commentary on this article by Moore, 28, 343–345



Environmental condition

So, we developed a mechanistic model of photosynthetic acclimation

Based on **optimization** and the **first principles** of plant physiological theory

Tansley review | [Free Access](#)

Eco-evolutionary optimality as a means to improve vegetation and land-surface models

Sandy P. Harrison , Wolfgang Cramer, Oskar Franklin, Iain Colin Prentice, Han Wang, Åke Brännström, Hugo de Boer, Ulf Dieckmann, Jaideep Joshi, Trevor F. Keenan, Aliénor Lavergne, Stefano Manzoni, Giulia Mengoli, Catherine Morfopoulos, Josep Peñuelas, Stephan Pietsch, Karin T. Rebel, Youngryel Ryu, Nicholas G. Smith, Benjamin D. Stocker, Ian J. Wright ... [See fewer authors](#) ^

First published: 15 June 2021 | <https://doi.org/10.1111/nph.17558>

Optimization: Least cost theory

Optimally, plants will maintain fastest rate of photosynthesis at the lowest summed resource cost (water and nutrient use)

Optimal photosynthesis

Photosynthesis = $f\{\text{stomatal conductance,}$
 $\text{photosynthetic biochemistry}\}$

Optimal photosynthesis

Photosynthesis = $f\{\text{stomatal conductance, photosynthetic biochemistry}\}$



Must predict optimal rates of both

Optimal photosynthesis

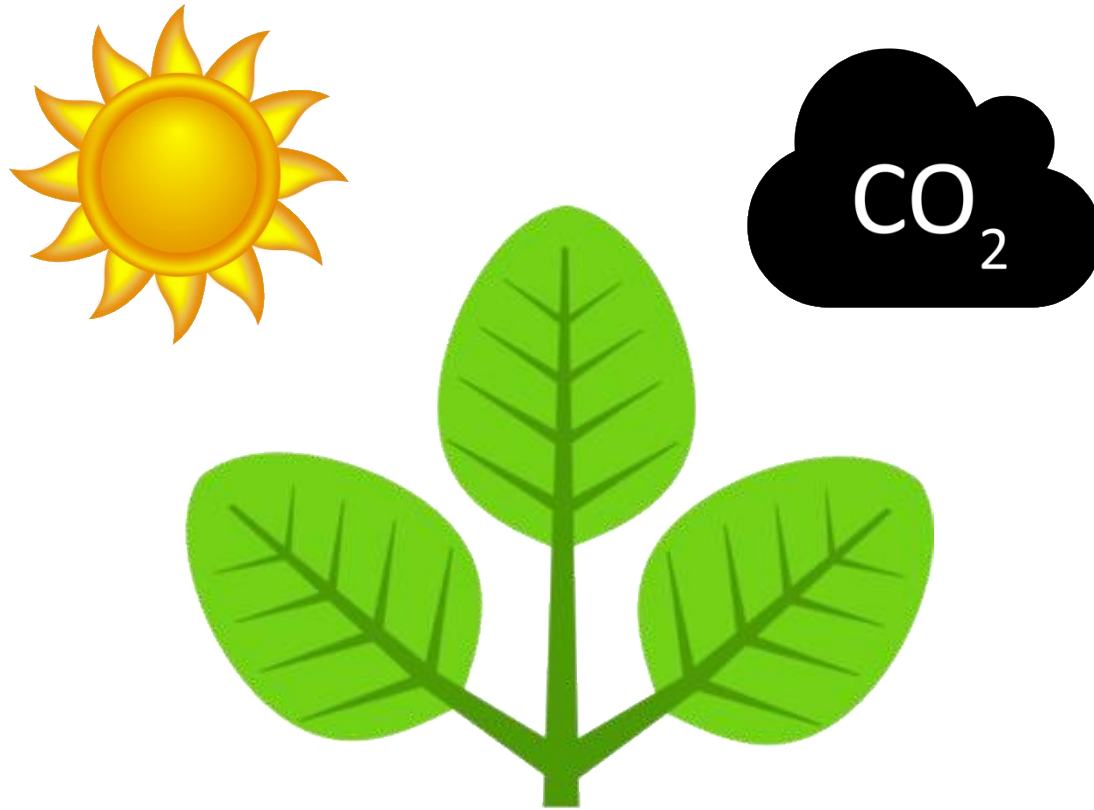
Photosynthesis = $f\{\text{stomatal conductance, photosynthetic biochemistry}\}$

[TALK TO ME LATER ABOUT THIS IF YOU ARE INTERESTED]

Optimal photosynthesis

Photosynthesis = f {stomatal conductance,
photosynthetic biochemistry}

Biochemistry optimization

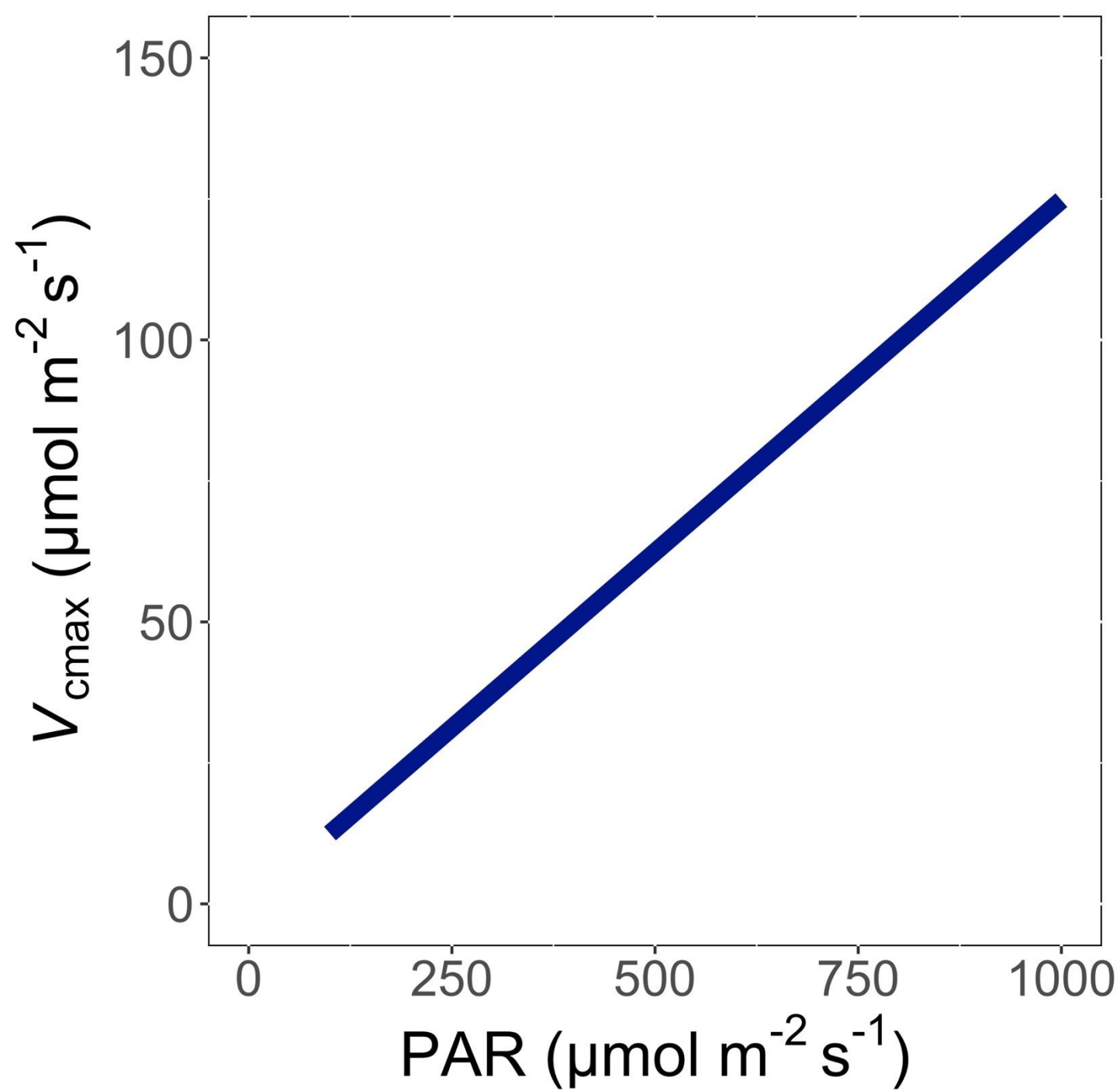


Plant
biochemistry
setup will aim
for equal
limitation by all
factors

[TALK TO ME LATER ABOUT THE MATHS]

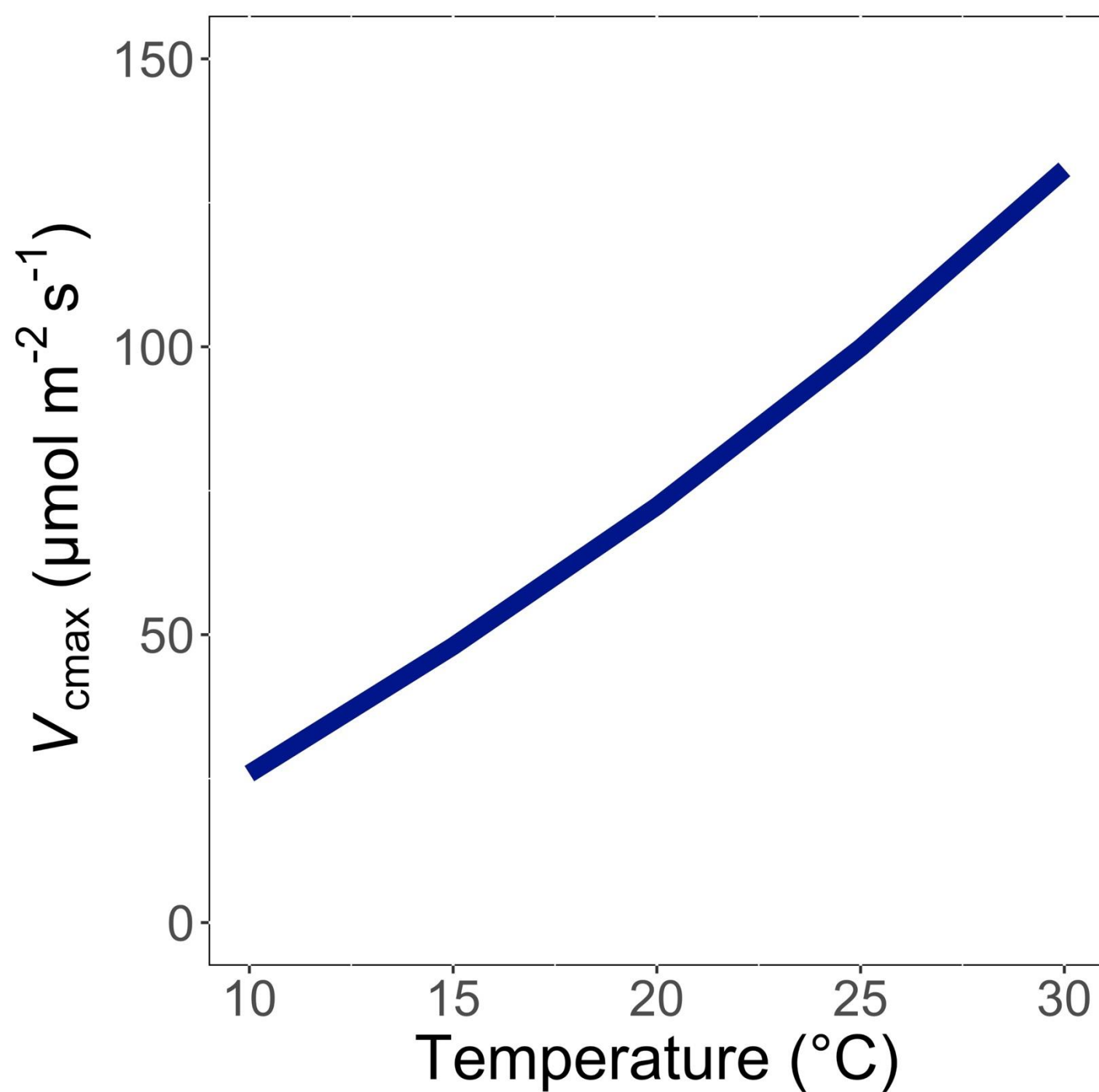
Biochemistry trait

The maximum rate of Rubisco carboxylation (V_{cmax})



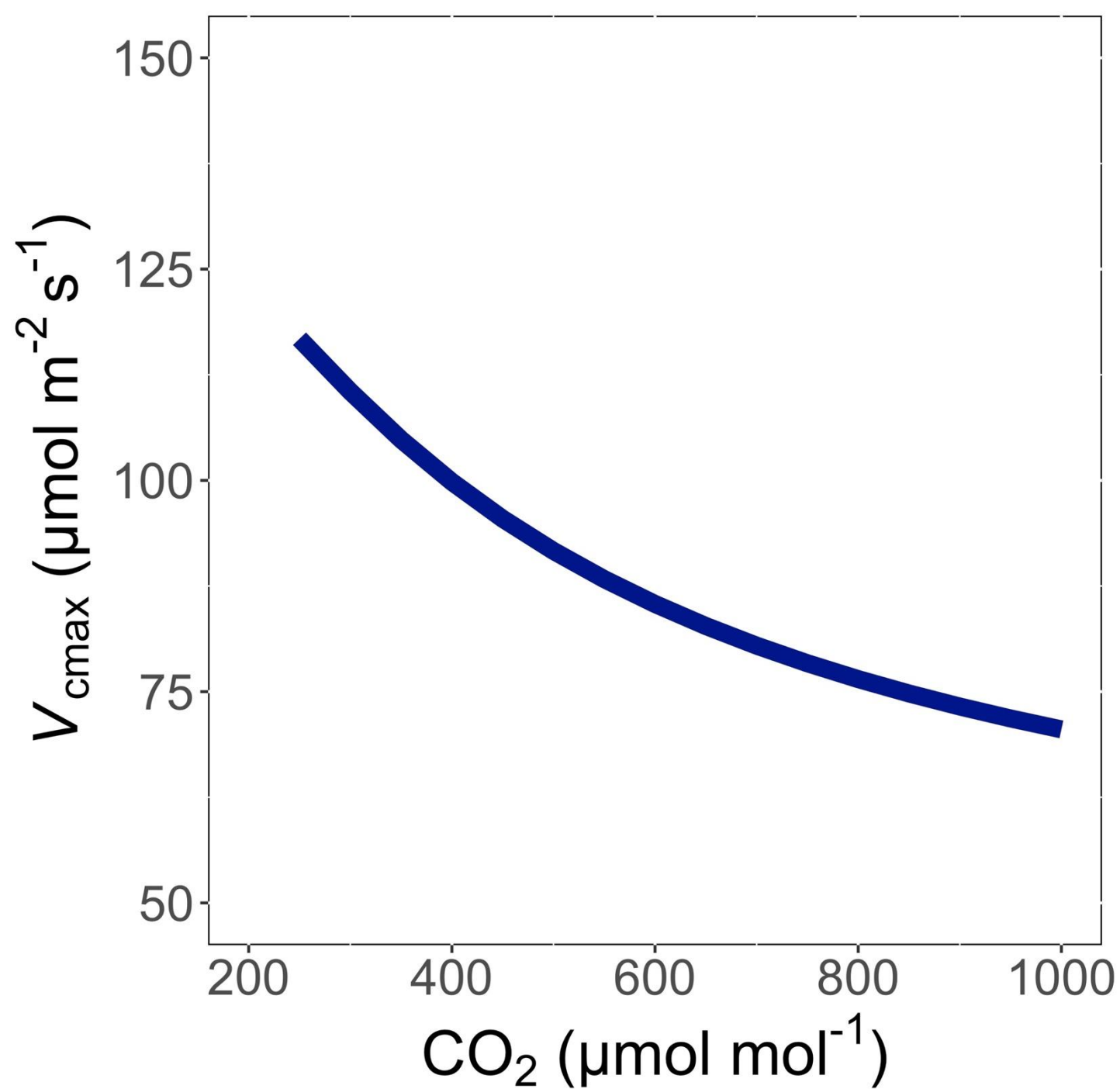
V_{cmax} increases
with light because
of greater electron
transport

Smith et al. (2019) *Ecol Letters*; Smith
and Keenan (2020) *Glob Change Biology*



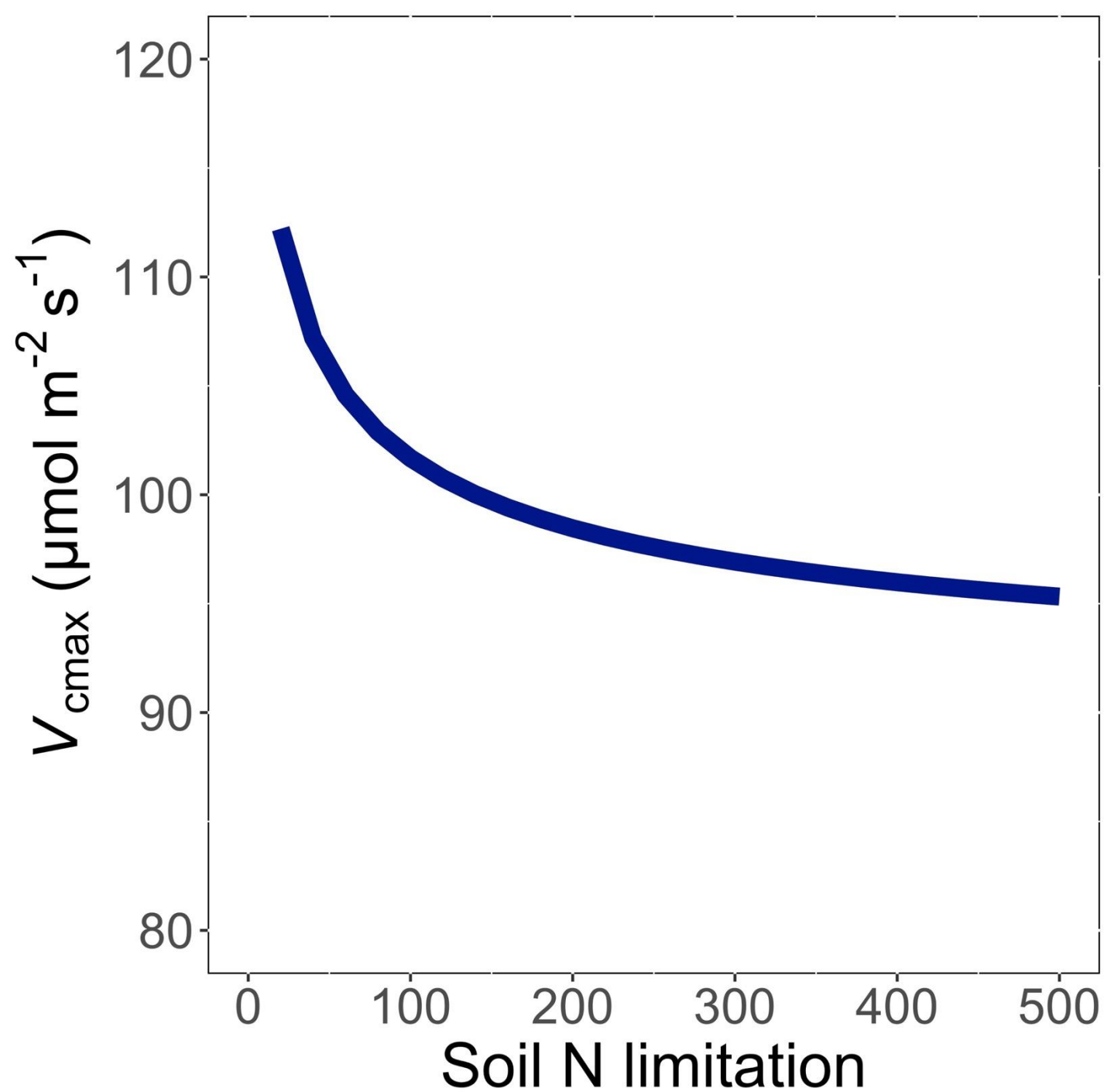
V_{cmax} increases
with temperature
because of greater
electron transport
and
photorespiration

Smith et al. (2019) *Ecol Letters*; Smith
and Keenan (2020) *Glob Change Biology*



V_{cmax} decreases
with CO_2 because
of greater CO_2 in
the leaf and less
photorespiration

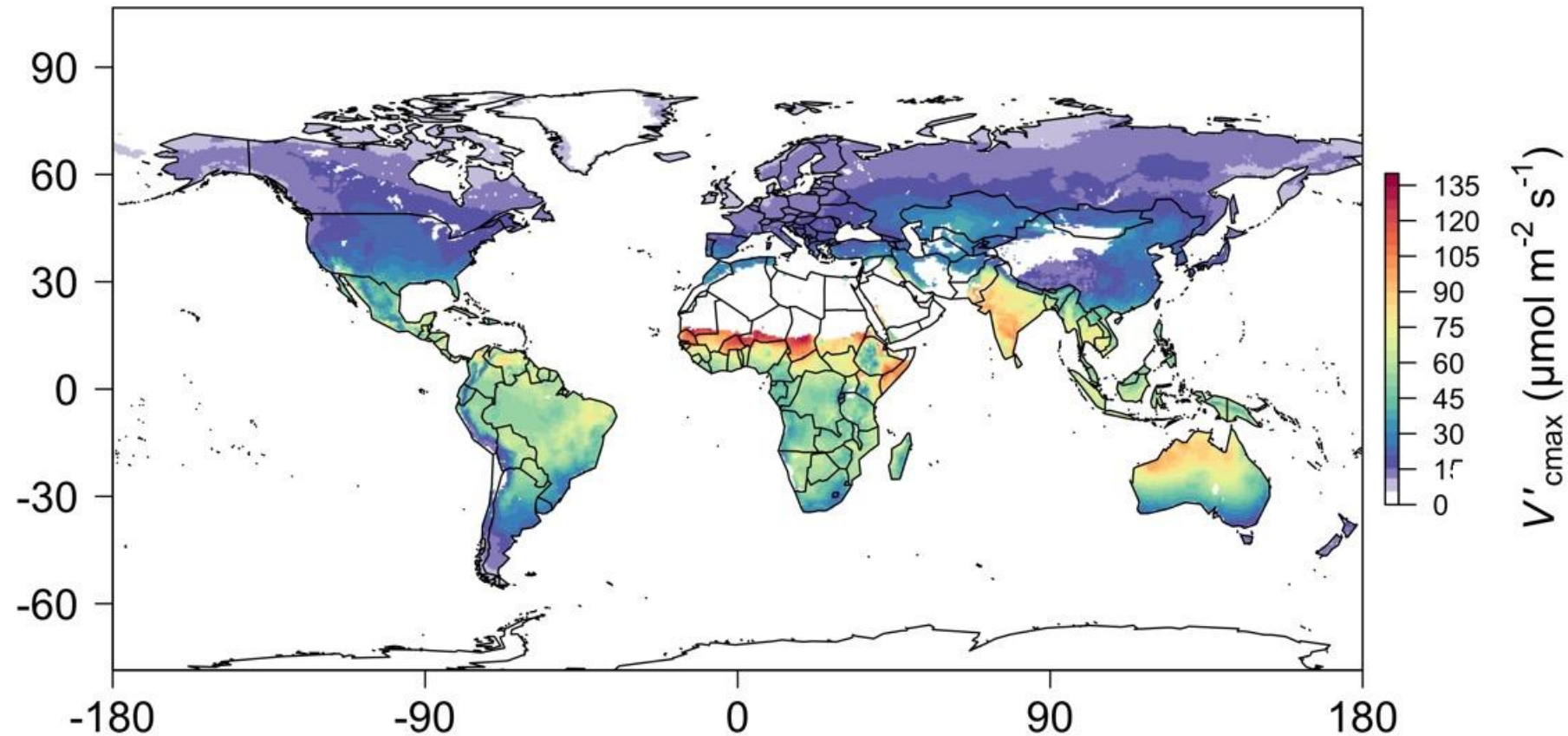
Smith et al. (2019) *Ecol Letters*; Smith
and Keenan (2020) *Glob Change Biology*



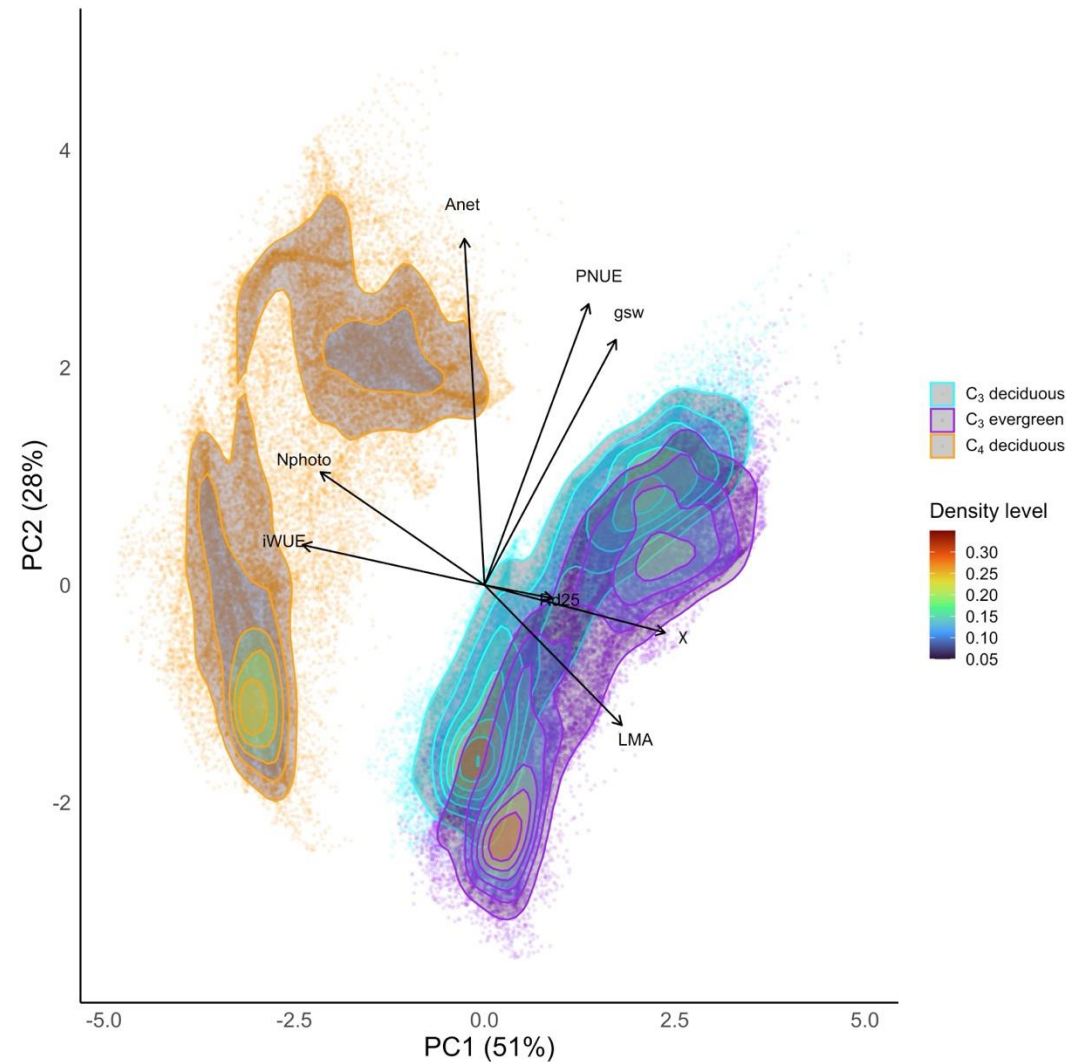
V_{cmax} decreases
with soil N
limitation because
Rubisco requires a
lot of N

Smith et al. (2019) *Ecol Letters*; Smith
and Keenan (2020) *Glob Change Biology*

We can predict optimal traits in different environments



We can even predict optimal trait strategies



Ok, great, but now what?

We can use the theory as a null model to explore acclimation mechanisms

Let's tackle some big questions in
plant ecophysiology!

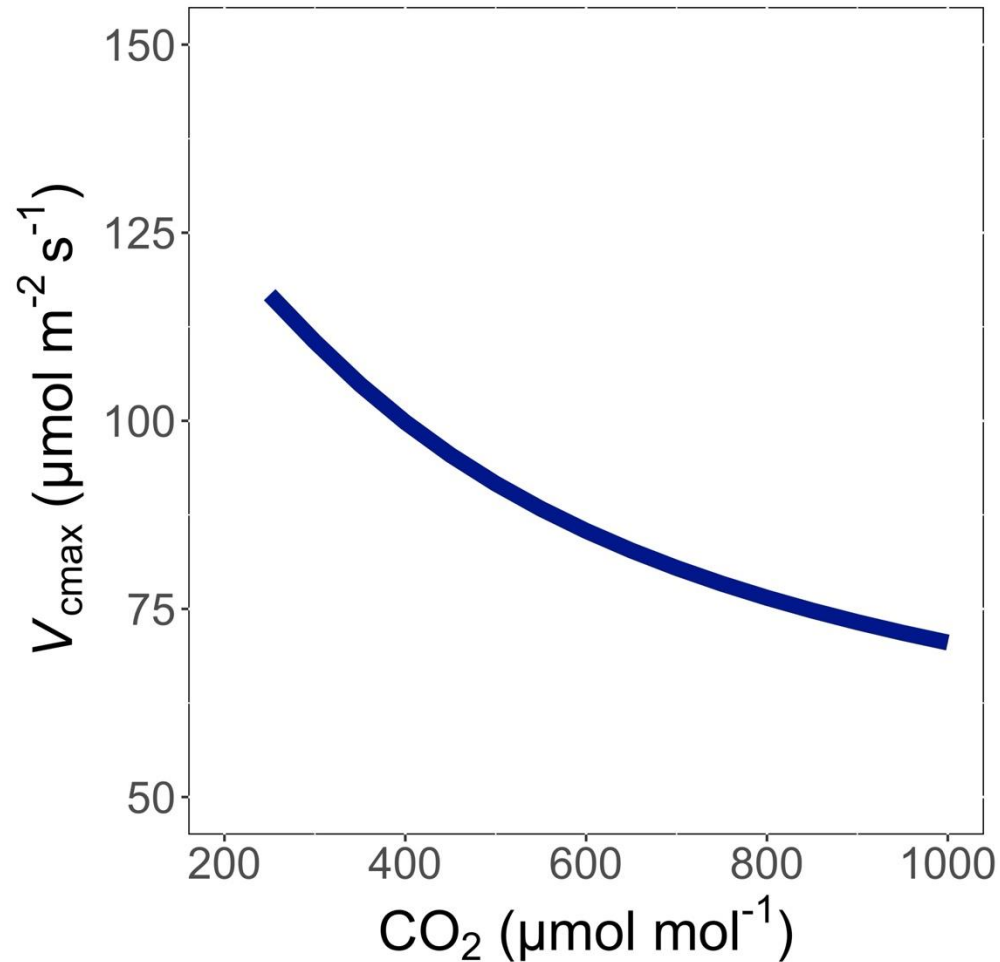
Big questions

1. Is photosynthesis **optimized** to the environment?
2. How do plants acclimate to **soil nitrogen**?
3. What does acclimation mean for **future** terrestrial biogeochemical cycling?
4. When is **C₄ photosynthesis** an advantage over C₃ photosynthesis?

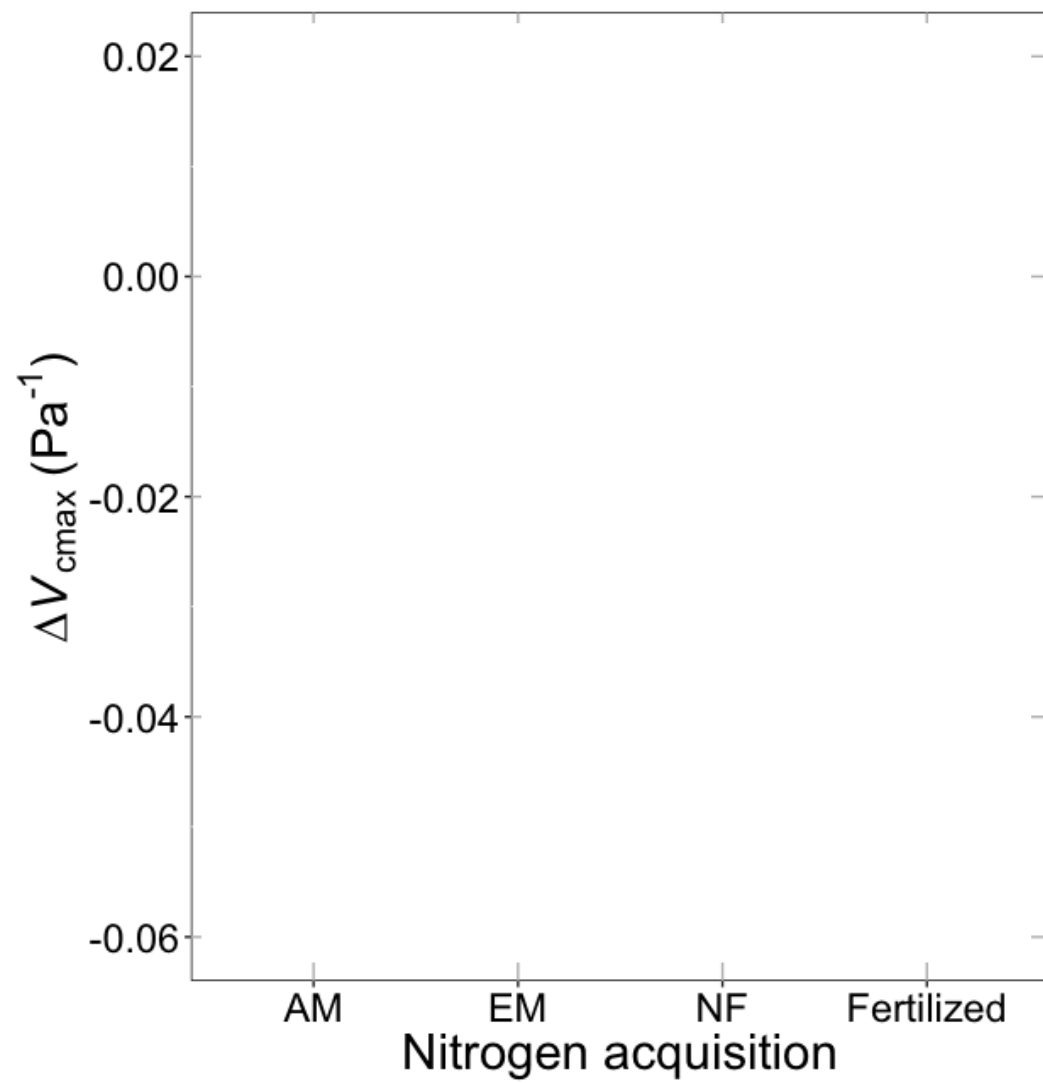
Big questions

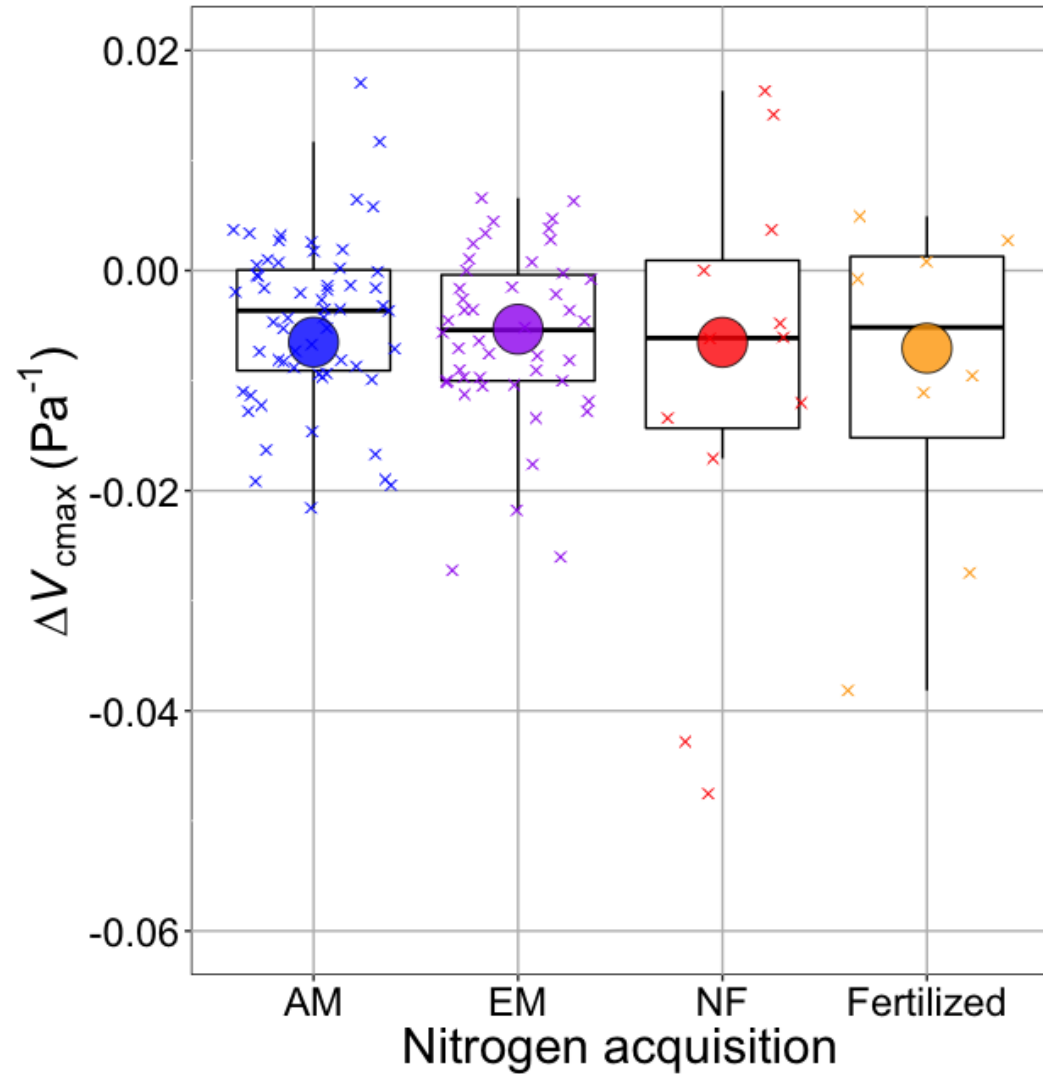
- ~~1. Is photosynthesis optimized to the environment?~~
- ~~2. How do plants acclimate to soil nitrogen?~~
3. What does acclimation mean for **future** terrestrial biogeochemical cycling? **A CO₂ story**
- ~~4. When is C₄ photosynthesis an advantage over C₃ photosynthesis?~~

Expected future responses



V_{cmax} decreases
with CO_2 because
of greater CO_2 in
the leaf and less
photorespiration

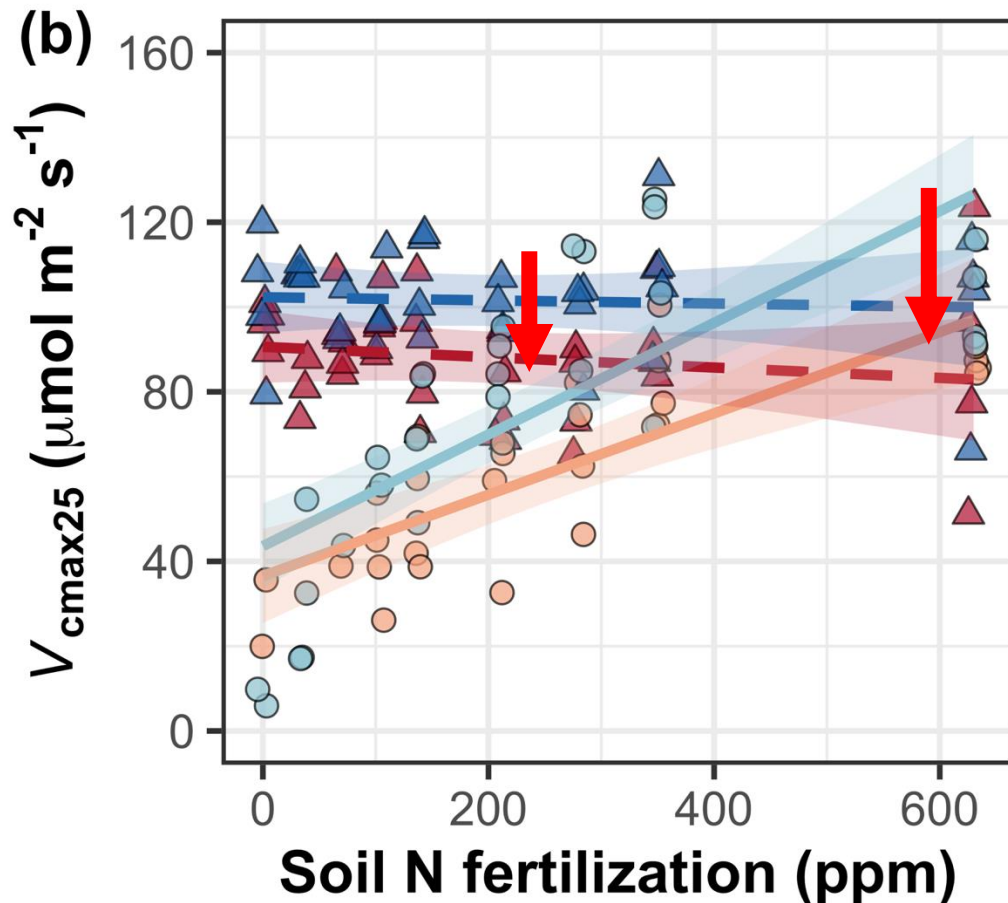




V_{cmax} changes with CO_2 in ways expected from optimization

Boxes = data = -0.0063 Pa^{-1}

Circles = predicted = -0.0066 Pa^{-1}



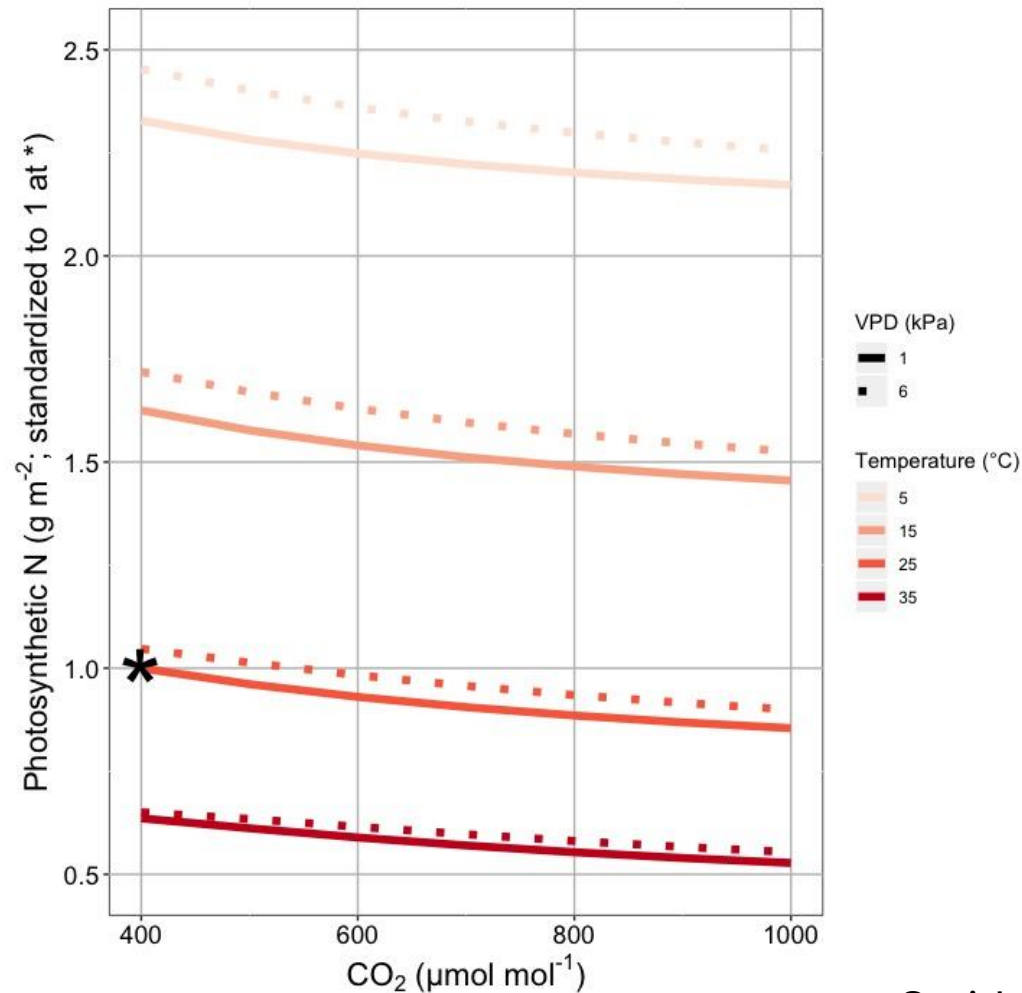
We further confirmed this universality of this response under 18 nitrogen by inoculation treatments

Treatment

- ▲ Elevated CO₂, inoculated
- Elevated CO₂, uninoculated
- ▲ Ambient CO₂, inoculated
- Ambient CO₂, uninoculated



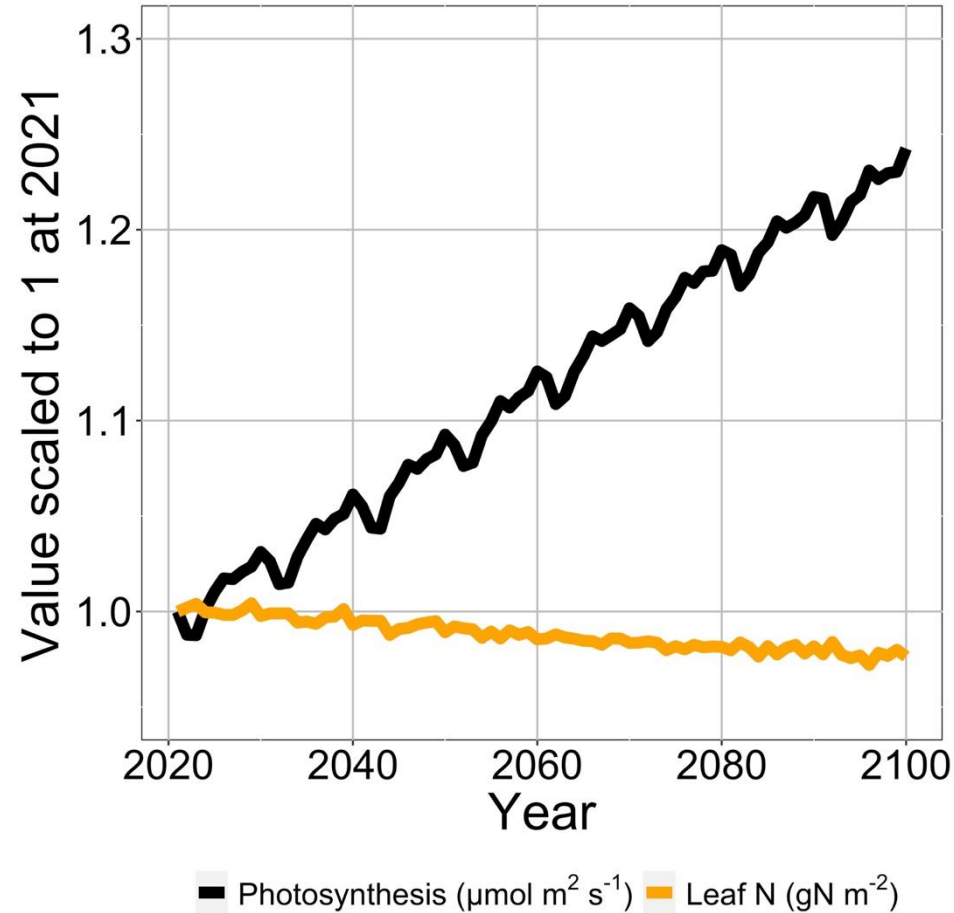
This generally suggests lower nitrogen demand under future conditions



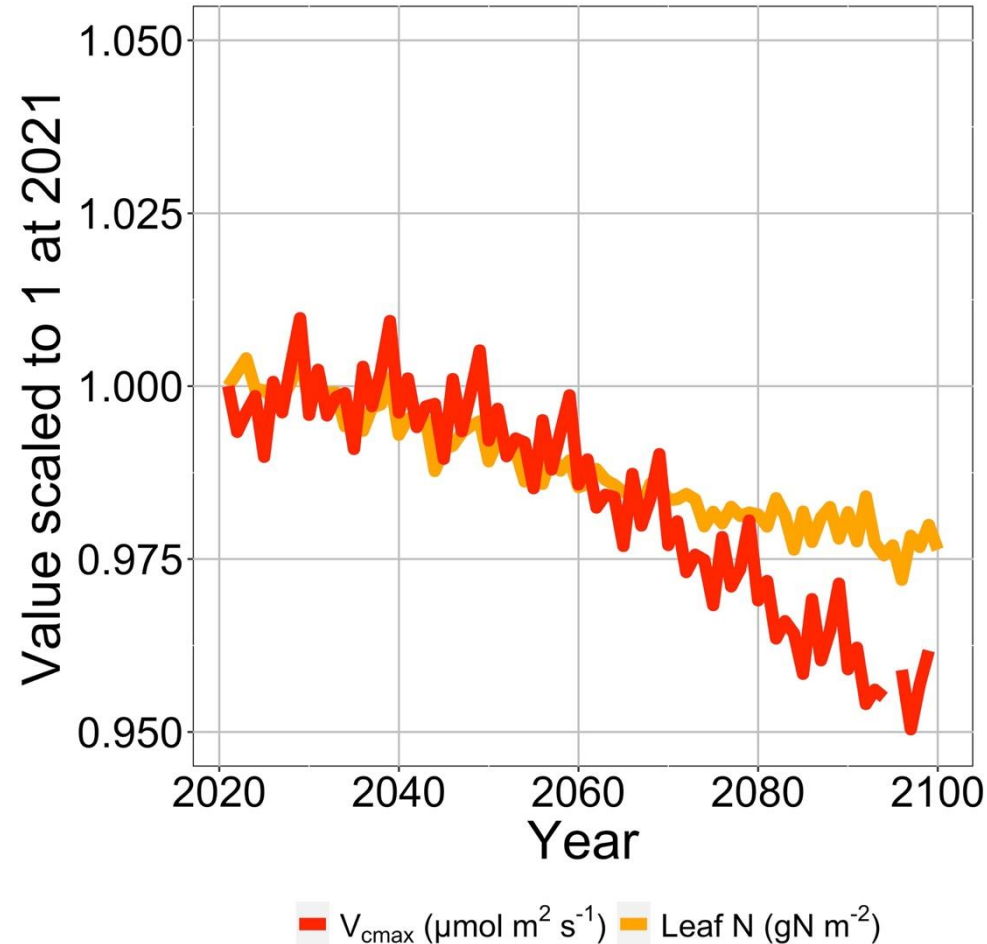
Let's run a model out into the future!



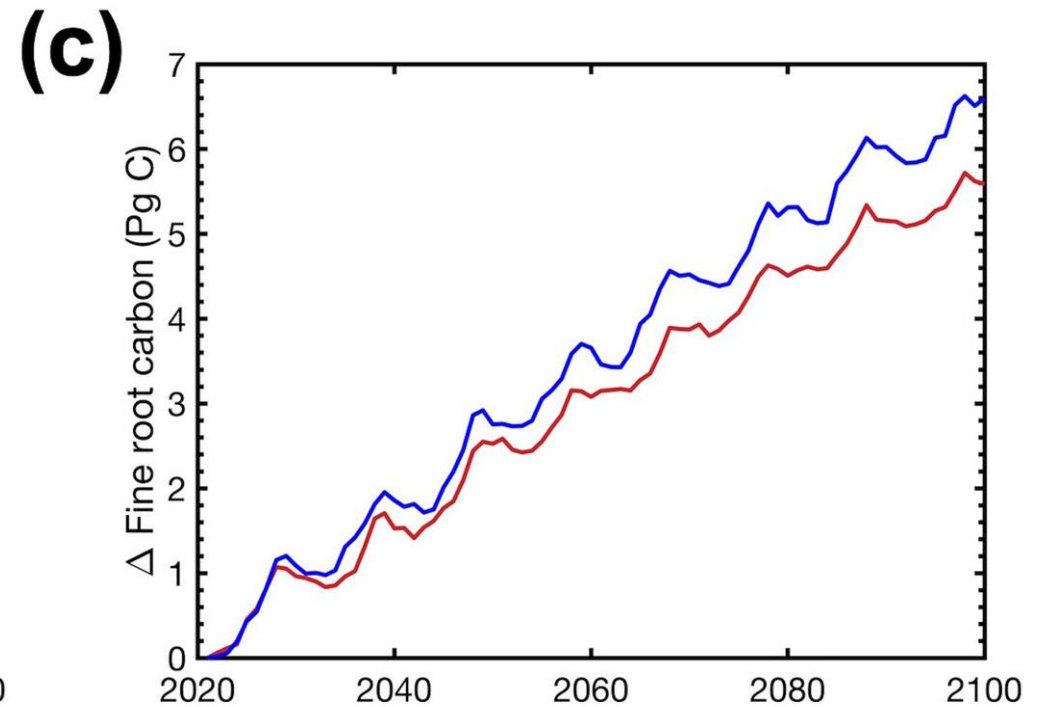
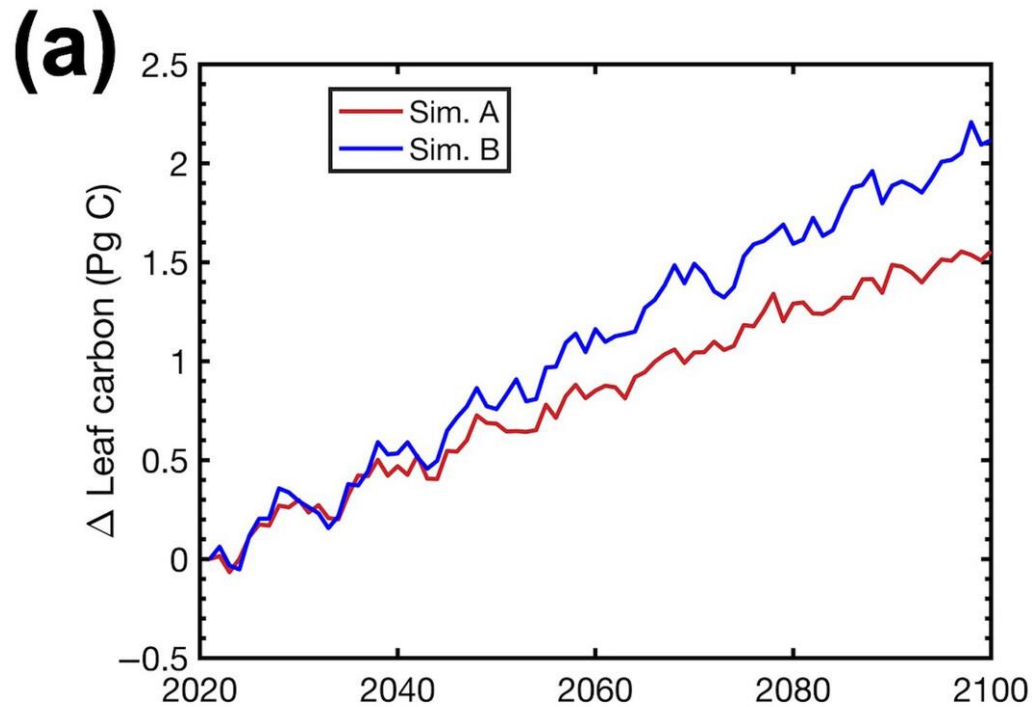
Photosynthesis increases with elevated CO₂ at lower leaf N



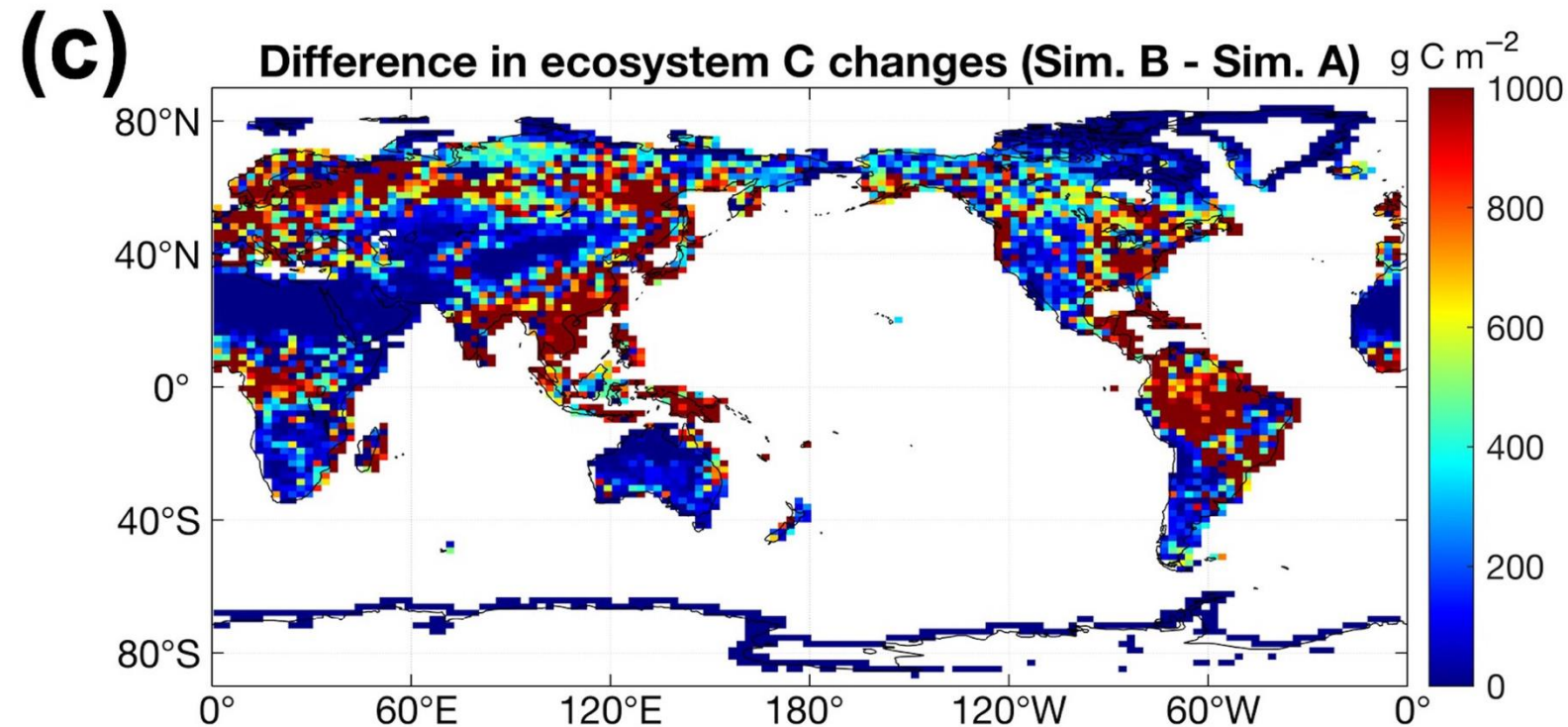
Leaf N reduction is due to a reduction in photosynthetic capacity



Plants build more leaves and fine roots

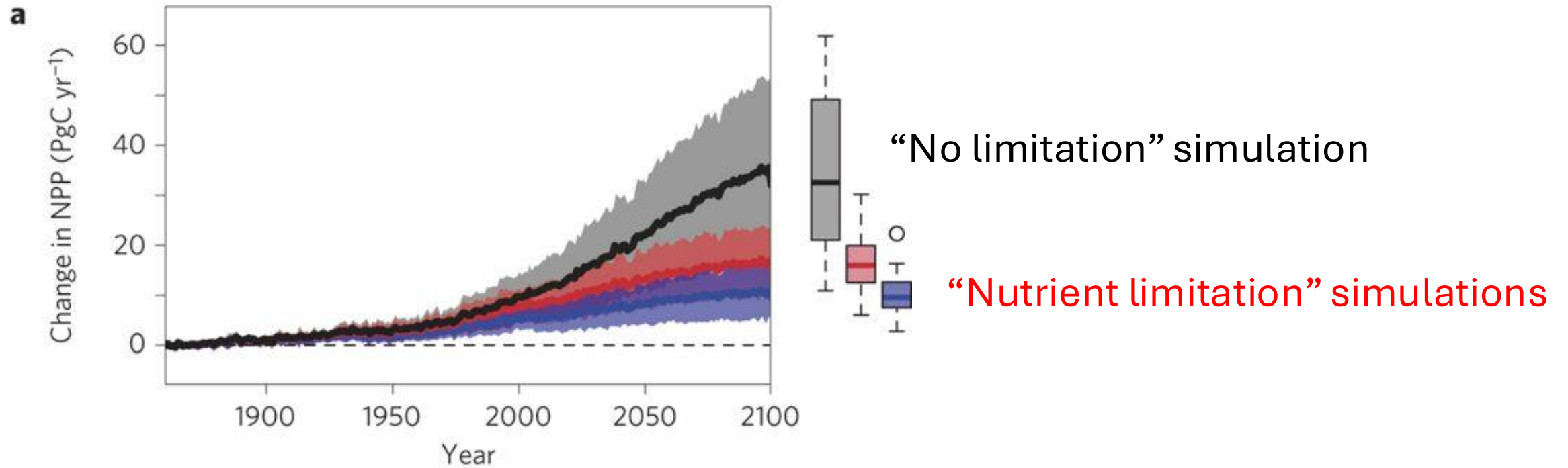


Leaf N savings increases long-term ecosystem carbon stocks



50% increase in
ecosystem carbon
from CO₂
acclimation

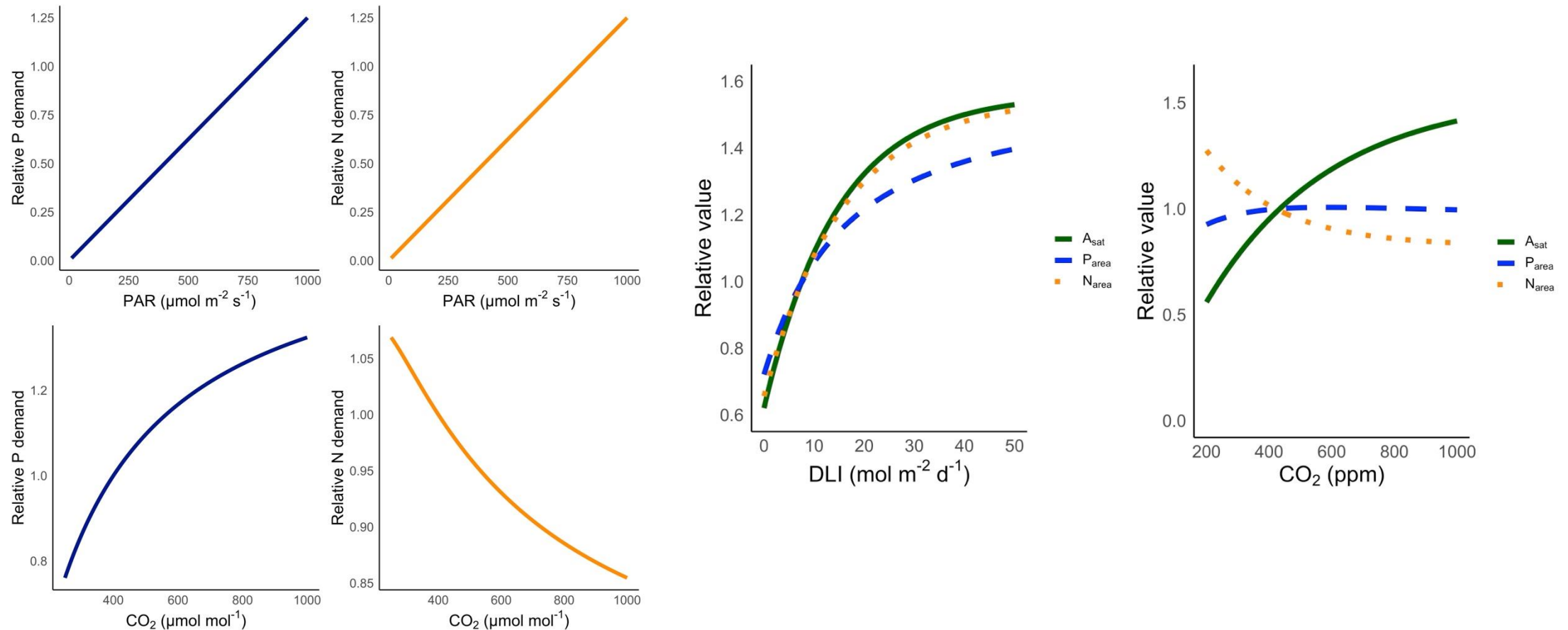
Need to rethink nutrient limitation in models?



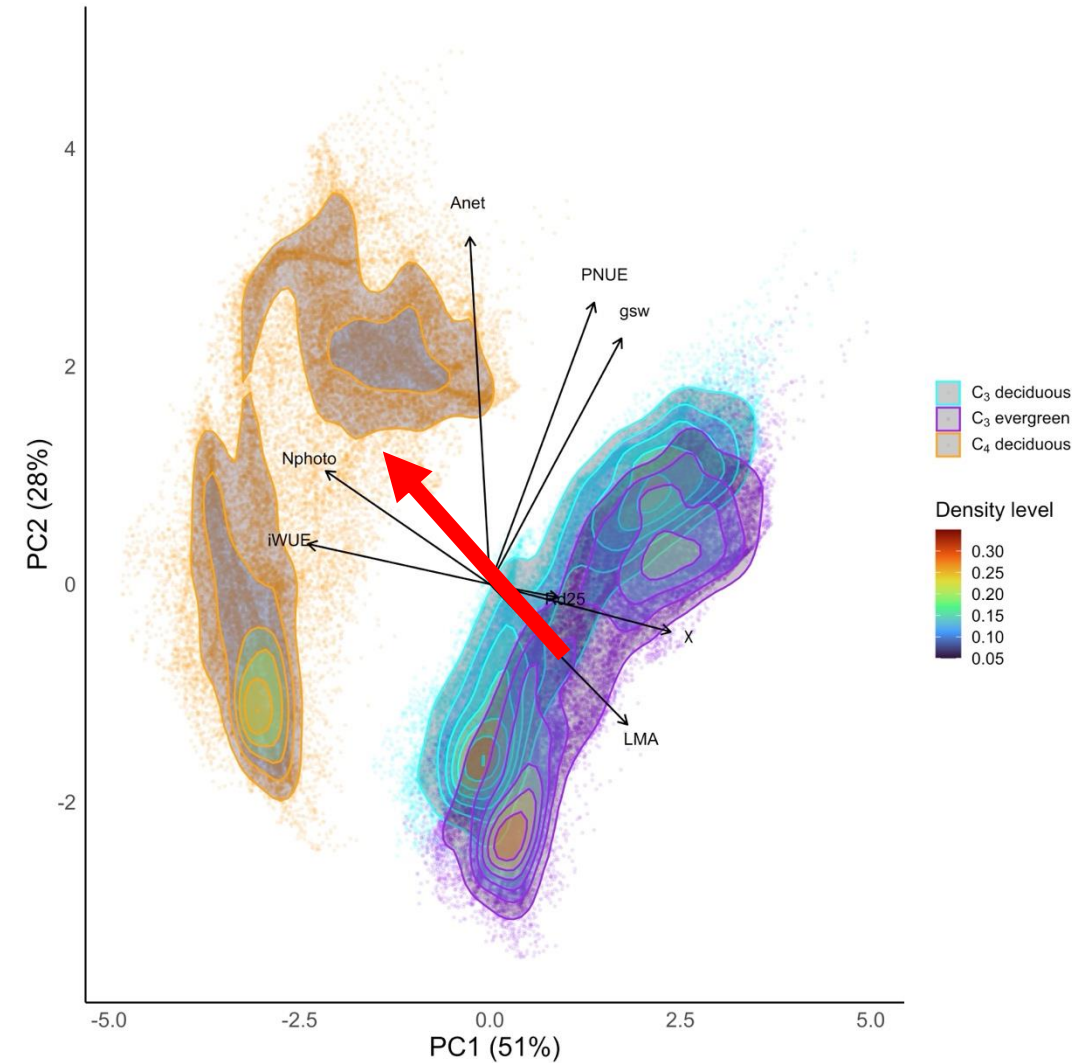
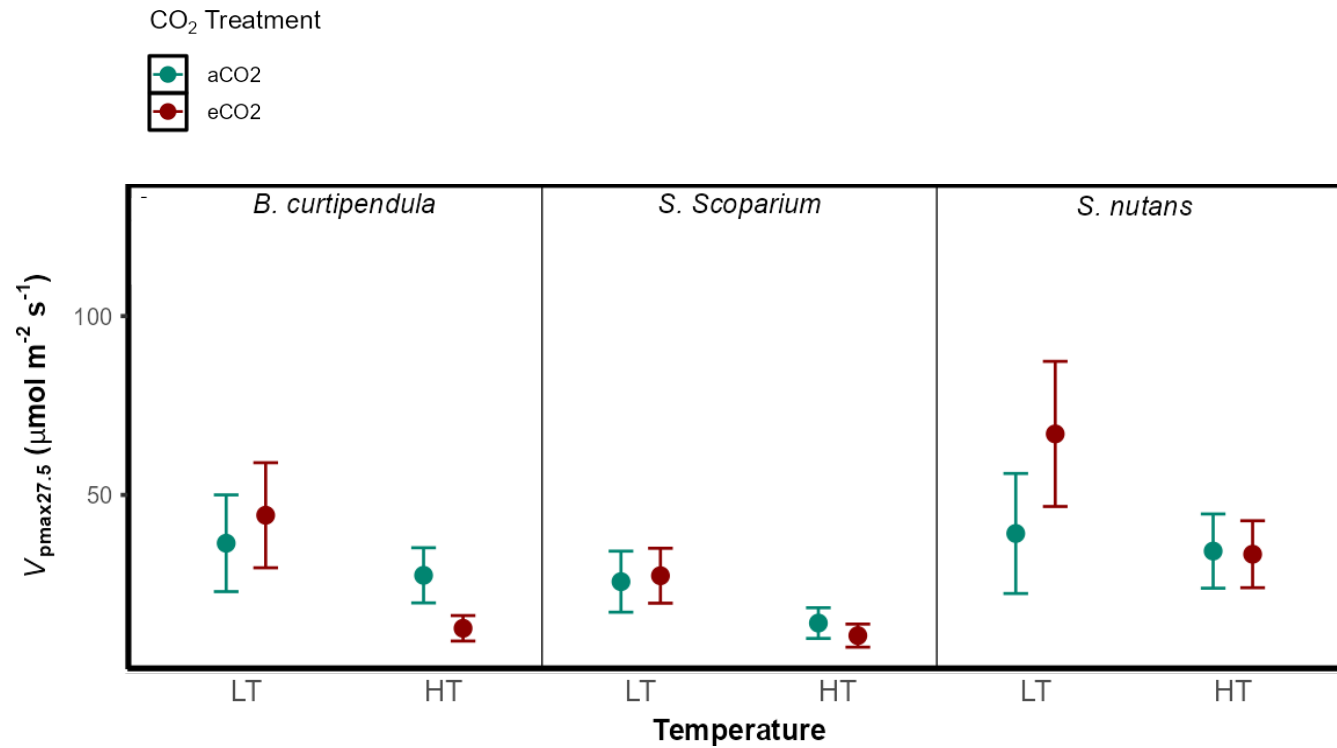
What does acclimation mean for future terrestrial biogeochemical cycling?

Photosynthesis will increase and per-leaf-area nutrient use will decrease, reducing N limitation and increasing future ecosystem C

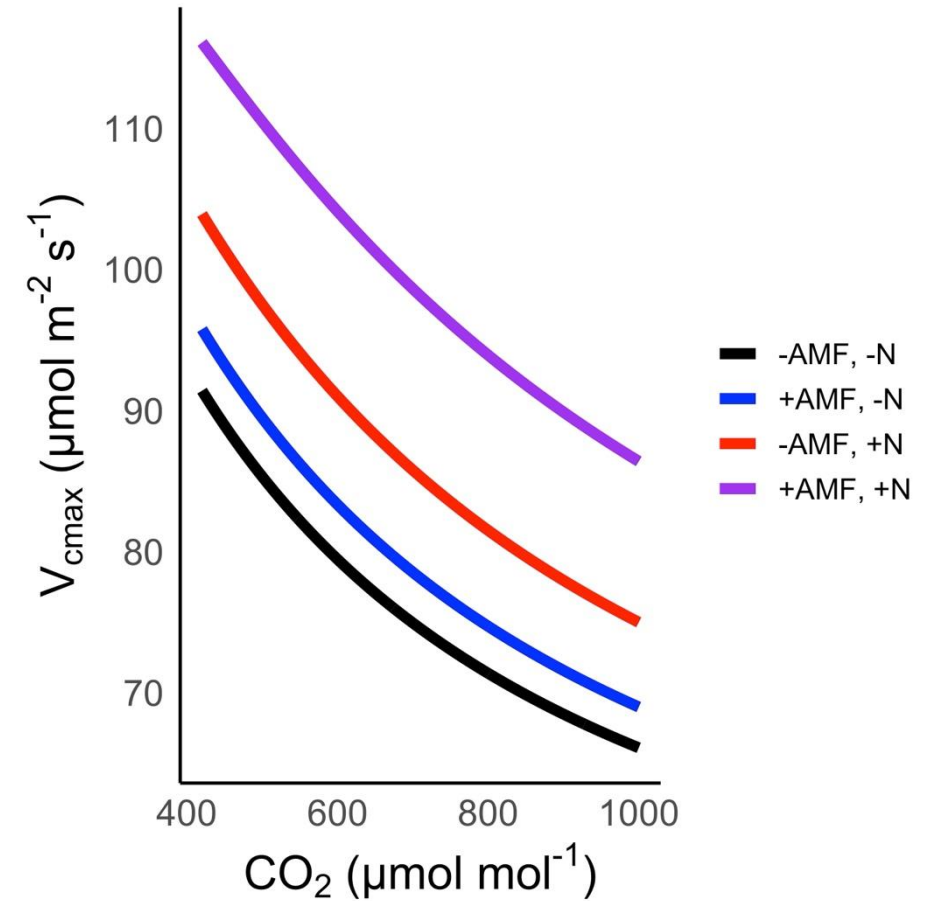
Looking forward: a simple theory for understanding photosynthesis-phosphorus relationships



Looking forward: integrating C₄ optimization



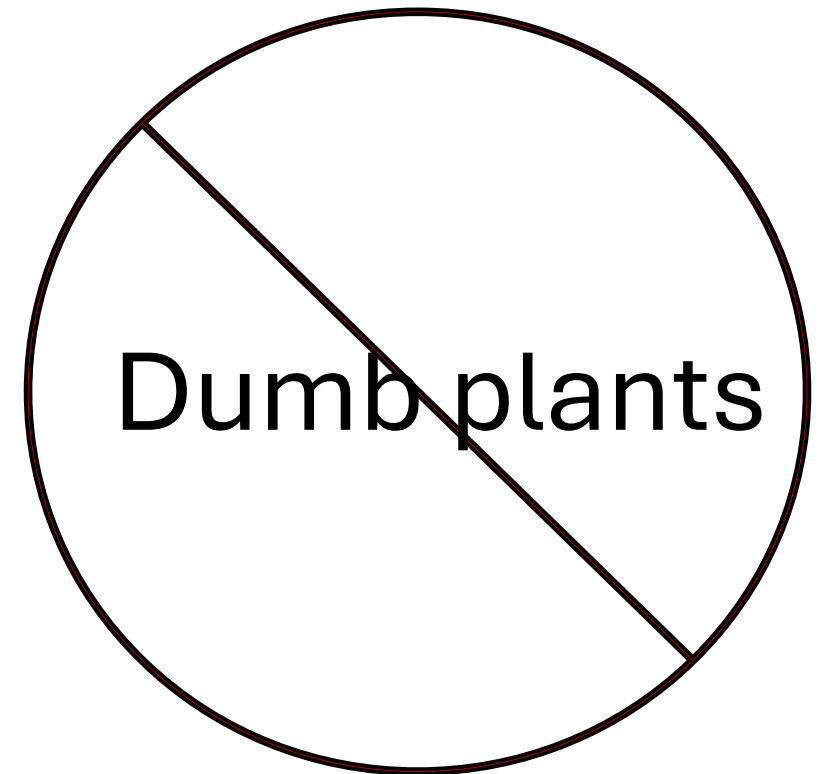
Looking forward: mycorrhizae-CO₂ interactions



Conclusions

Conclusions

- Plants aren't dumb!
 - Assuming plants don't dynamically respond to their environment can lead to poor understanding of ecological functioning from individuals to the Earth system



Conclusions

Quantified physiological theory can:

1. Improve mechanistic understanding of physiological processes that underlie higher level responses
2. Produce more reliable future projections



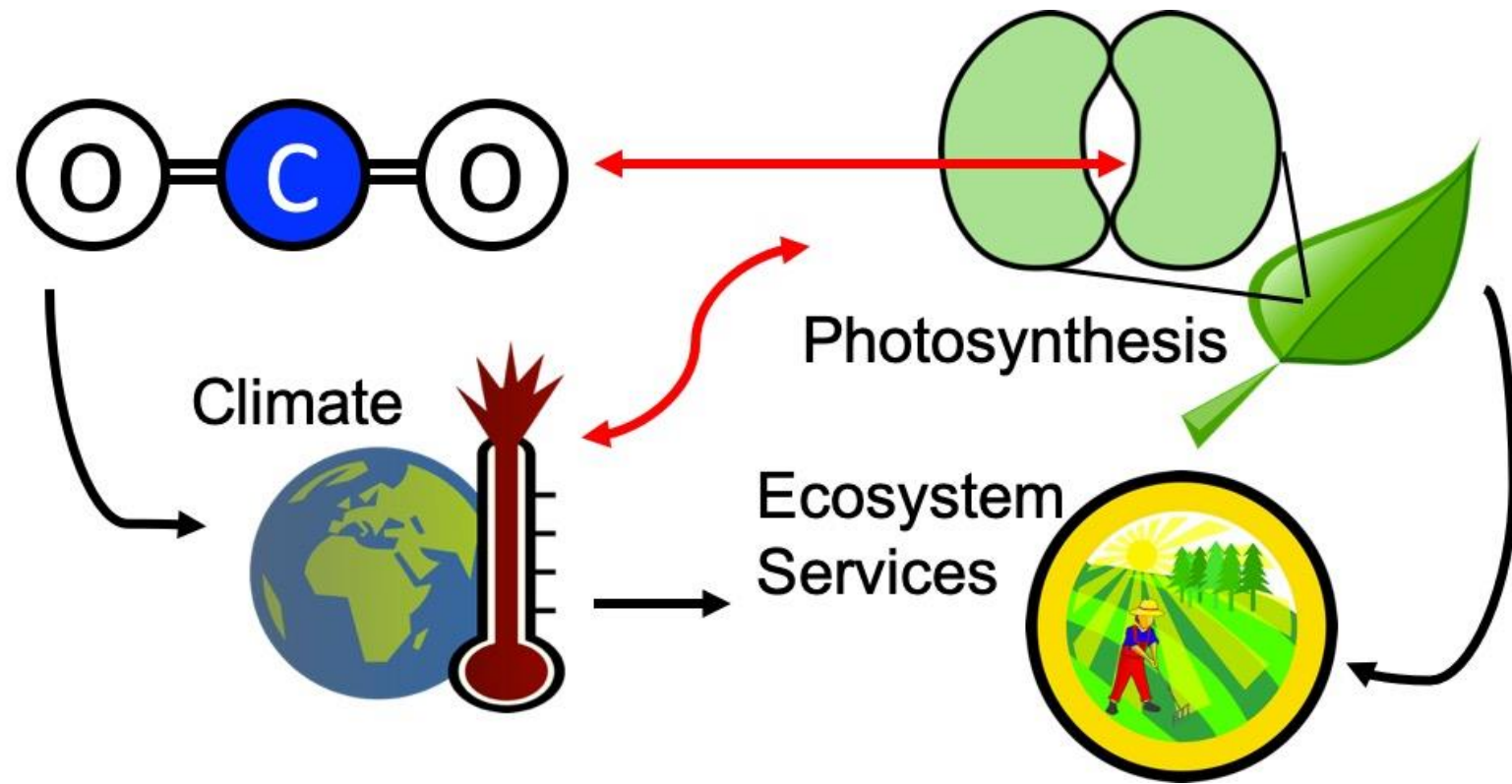
$f(x)$



$f(x)$



And ultimately lead to a better understanding of how future ecosystem services will be impacted by global change

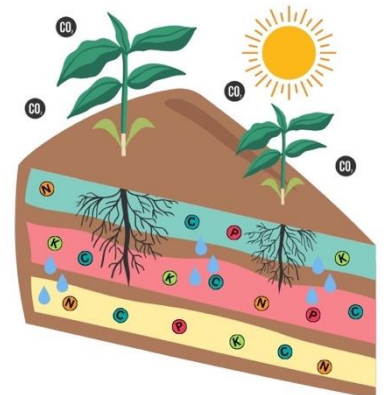


Quick notice

- Lab is moving to the University of Nebraska - Lincoln
- Starting Fall 2026
- Reach out if you are interested in collaborating



UNIVERSITY *of* NEBRASKA
LINCOLN



Presentation available at: www.github.com/SmithEcophysLab/seminar/

Contact: nick.smith@ttu.edu, nsmith87@unl.edu



Thanks!